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Paper 53
Date: July 12, 2022

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

ALZHEON INC.,
Petitioner,

v.

RISEN (SUZHOU) PHARMA TECH CO., LTD,
Patent Owner.

IPR2021-00347
Patent 10,472,323 B2

Before ERICA A. FRANKLIN, JOHN G. NEW, and ZHENYU YANG, *Administrative Patent Judges.*

NEW, *Administrative Patent Judge*

JUDGMENT
Final Written Decision
Determining All Challenged Claims Unpatentable
Denying Petitioner's Motion to Strike
Denying Patent Owner's Motion to Amend
35 U.S.C. § 318(a)

I. INTRODUCTION

We have jurisdiction to hear this *inter partes* review under 35 U.S.C. § 6, and this Final Written Decision is issued pursuant to 35 U.S.C. § 318(a) and 37 C.F.R. § 42.73. For the reasons set forth below, we determine that Alzheon Inc. ("Petitioner") has established by a preponderance of the evidence that claims 1–20 of U.S. Patent No. 10,472,323 B2 (Ex. 1001, "'323 Patent") are unpatentable. Furthermore, because we similarly find that Petitioner has established by a preponderance of the evidence that proposed substitute claims 21–23 of Risen (Suzhou) Pharma Tech Co., Ltd's ("Patent Owner") Contingent Motion to Amend (Paper 19) are unpatentable, we deny the Contingent Motion to Amend.

A. Procedural History

On December 18, 2020, Petitioner filed a Petition (Paper 1, "Petition") seeking *inter partes* review of claims 1–20 of the '323 Patent. Patent Owner timely filed a Preliminary Response. (Paper 9). Pursuant to 35 U.S.C. § 314, we instituted trial on July 14, 2021, upon all of the challenged claims of the '323 Patent (Paper 10, "Institution Decision" or "Dec.").

After institution of trial, Patent Owner filed a Response (Paper 18, "PO Resp."), to which Petitioner filed a Reply (Paper 34, "Pet. Reply"), and Patent Owner, in turn, filed a Sur-reply (Paper 40, "Sur-reply").

Patent Owner also filed a Contingent Motion to Amend (Paper 19, "Mot. Amend"). In response to Patent Owner's request (*see* Mot. Amend 1), on March 24, 2002, we issued Preliminary Guidance (Paper 39) on the Contingent Motion to Amend. Petitioner filed an Opposition to the

Contingent Motion to Amend (Paper 35, "Opp."), to which Patent Owner filed a Reply (Paper 41, "Opp. Reply"), and Petitioner filed a Sur-Reply (Paper 42, "Opp. Sur-Reply").

Petitioner relies upon a Declaration of Dr. F. Peter Guengerich, (Ex. 1002). Patent Owner originally relied upon a Declaration by Dr. Steven Wagner (Ex. 2068). Regrettably, Dr. Wagner passed away on March 12, 2022 and, prior to that, his failing health prevented his deposition by Petitioner. Patent Owner therefore requested that Dr. S. David Kimball be allowed to substitute testimony for the ailing Dr. Wagner. *See* Ex. 3002. Following a telephone conference on February 1, 2022, we ordered that Dr. Kimball be allowed to substitute a Declaration (Paper 2099) in place of Dr. Wagner's Declaration, and give testimony under deposition, on the condition that Dr. Kimball adopt Dr. Wagner's Declaration with no changes to its content except for the relevant biographical details. Paper 26, 2–3. Patent Owner also relies on a Declaration of Dr. Jonathan L. Sessler (Ex. 2042).

Oral argument was held on May 5, 2022. Subsequent to requests by both parties (Papers 50, 51), additional time for oral argument was allowed to both Petitioner and Patent Owner under the Legal Experience and Advancement Program (LEAP). A transcript of the oral argument is included in the record. (Paper 52, "Hearing Trans.").

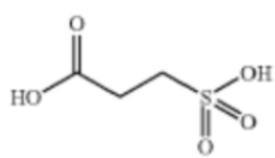
B. Related Proceedings

The parties agree that there are no related matters concerning the '323 Patent. Pet. 5; Paper 4, 2.

C. The ’323 Patent (Ex. 1001)

The ’323 patent is directed to isotope-enriched 3-amino-1 propanesulfonic acid (“3APS” or “tramiprosate”) and derivatives, compositions thereof, and methods of using them in therapeutic applications, such as in the prevention and treatment of Alzheimer’s disease (“AD”). Ex. 1001, col. 1, ll. 15–19. Tramiprosate is a small molecule drug that was developed as a treatment for symptoms of AD and was known in the prior art. *Id.* at col. 1, ll. 47–51. Tramiprosate is believed to act by reducing amyloid aggregation, deposition, and/or the load of amyloid in the brain through its binding to soluble amyloid-β (“Aβ”) peptide. *Id.* at col. 1, ll. 59–61.

Tramiprosate is extensively metabolized *in vivo* to produce three potential metabolites: 2-carboxyethanesulfonic acid, 3-hydroxy-1-propanesulfonic acid, and 3-acetylamino-1-propansulfonic acid. *Id.* at col. 1, ll. 63–66. The only major metabolite of tramiprosate produced in mice, rats, dogs, and humans is 2-carboxyethanesulfonic acid (“2-CES”), which is depicted below:

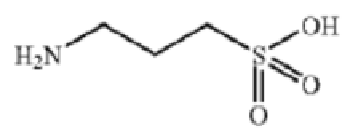
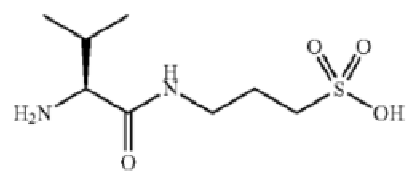


2-carboxyethanesulfonic acid

Structure of 2-carboxyethanesulfonic acid, the sole metabolite of tramiprosate in mammals

Ex. 1005, col. 2, ll. 15–17, 25–30. This metabolism of tramiprosate has significant effect on its pharmacokinetic profile and, consequently, its pharmaceutical efficacy. Ex. 1001, col. 2, ll. 1–3. One approach to improving the therapeutic effectiveness of tramiprosate in the face of such metabolism is *via* the use of prodrugs and derivatives of tramiprosate that will generate tramiprosate *in vivo* after administration to a subject. *Id.* at col. 2, ll. 6–11 (citing and incorporating by reference in its entirety Ex. 1005).

A preferred prodrug of tramiprosate, L-valyl-3-aminopropanesulfonate ("Val-APS") is expressly disclosed by the '323 patent. Ex. 1001, Table 1, comp. 4. The structures of tramiprosate and Val-APS are depicted below:

	
Tramiprosate 3-amino-1-propanesulfonic acid (3APS)	L-valyl-3-aminopropanesulfonate (Val-APS)

Chemical structures of tramiprosate (left) and Val-APS (right)

See Pet. 9; *see also* Ex. 1005, col. 2, ll. 20–35.

Claims 1–20 of the '323 patent are directed to compounds that include isotopically-substituted tramiprosate where deuterium is substituted for hydrogen at the 3-carbon (e.g., claims 1 and 3)[1] and isotopically-substituted Val-APS where deuterium is substituted for hydrogen at corresponding α-carbon (e.g., claim 1). The claims also encompass optional or mandatory substitution of isotopes of carbon, nitrogen, or oxygen in Val-APS or tramiprosate.

[1] The "3-carbon" or "third carbon" is the carbon immediately adjacent to the amine moiety in tramiprosate and the corresponding carbon in Val-APS, and designated CR2 in claims 1 and 3.

D. Illustrative Claim

Independent claims 1 and 3 of the ’323 patent are representative and recite:

> 1. A compound of Formula (I), or a pharmaceutically acceptable salt or ester thereof:
>
> $$R^1R^2X–CR_2–CH_2–CH_2–SO_3H \qquad \text{(I)}$$
>
> where:
>
> R^1 and R^2 are independently a hydrogen of natural abundance or a protecting group, wherein said protecting group is isotope-enriched or non-isotope enriched, said protecting group being selected from acyl, carbonyl, thiocarbonyl, and carbamoyl groups, wherein at least one of R^1 and R^2 is a protecting group;
>
> X is a nitrogen of natural abundance or ^{15}N, or a combination thereof; and
>
> R is a hydrogen of natural abundance, a deuterium (D) or a combination thereof;
>
> provided that at least one of X, R, R^1 and R^2 comprises an atom that is not of natural abundance;
>
> provided that the compound is not N-acetyl-3-aminolpropanesulfonic acid.

Ex. 1001, col. 47, ll. 33–52.

> 3. A compound of Formula (II), or a pharmaceutically acceptable salt or ester thereof:
>
> where:
>
> $$H_2X–CR_2–CH_2–CH_2–SO_3H \qquad \text{(II)}$$

X is ^{15}N; and

R is a hydrogen of natural abundance, a deuterium (D) or a combination thereof.

Id. at col. 47, ll. 60–67. Claims 1 and 3 thus respectively include within their scopes the prodrug form of tramiprosate (including Val-APS) and tramiprosate.

E. Asserted Grounds of Unpatentability and Asserted Prior Art

Petitioner contends that claims 1–20 of the ’323 Patent are unpatentable based on the following grounds:

Claims Challenged	**35 U.S.C. §**	**References/Basis**
1–17, 19, 20	103	Kong[2], Czarnik[3]
1–20	103	Kong, Czarnik, Tolar[4]
3, 19	103	Czarnik, Kong

[2] Kong et al. (US 8,748,656 B2, June 10, 2014) (“Kong”) (Ex. 1005).

[3] Czarnik (US 2009/0076167 A1, March 19, 2009) (“Czarnik”) (Ex. 1007).

[4] Tolar et al. (US 10,471,029 B2, November 12, 2019) (“Tolar”) (Ex. 1008).

II. ANALYSIS

A. Petitioner's Motion to Exclude the Kimball Declaration

Before turning to our analysis proper of the patentability of claims 1–20 of the '323 patent, we address Petitioner's Motion to Exclude the Declaration of Patent Owner's expert witness, Dr. S. David Kimball ("Mot. Exclude"). Paper 43. As we have related *supra*, Dr. Kimball adopted the Declaration of Dr. Wagner after the latter's unfortunate demise. The substance of the Kimball Declaration is identical to the original Wagner Declaration except for the relevant biographical details. *See* Paper 26, 2–3. Deposition testimony of Dr. Kimball was taken remotely on February 24, 2022. Ex. 1069. Petitioner subsequently filed a Motion to Exclude the Declaration of Dr. Kimball on April 14, 2022.

In response to the Petitioner's Motion to Exclude, Patent Owner filed an Opposition to the Motion ("Opp. Exclude," Paper 45), to which Petitioner filed a Reply ("Reply Exclude," Paper 47).

1. Dr. Kimball's lead compound analysis

Petitioner first argues that, in his Declaration, Dr. Kimball applied the wrong legal standard for the lead compound analysis in the declaration. Mot. Exclude 5. According to Petitioner, Dr. Kimball's Declaration purports to address "whether a POSA would have selected tramiprosate, Val-APS, or compound 6 as a lead compound for further modification under the lead compound analysis," but Dr. Kimball admitted at his deposition that he was not applying the "legal lead compound test." *Id.* at 6 (citing (Kimball Decl. ¶ 36 and quoting Ex. 1069, p. 35, ll. 10–13). Rather, Petitioner asserts, Dr. Kimball applies his own factors, e.g., taking the position that to have a lead compound "any POSA would have to come forward with a disease-modifying treatment for Alzheimer's. Treating symptoms is simply not adequate." *Id.* (quoting Ex. 1069, p. 30, ll. 7–13, p. 123, ll. 14–24).

Petitioner further contends that Dr. Kimball also required a lead compound "have a mechanism of action" and "a therapeutic rationale" and that "the POSA actually demonstrates that there's credible literature supporting those." *Id.* (quoting Ex. 1069, p. 33, ll. 7–13). Petitioner contends that Dr. Kimball thus imposed an unduly high standard that is inconsistent with the law, applying an improper methodology that cannot actually determine whether the compounds are lead compounds under the controlling case law. *Id.*

Patent Owner responds that, at the time the '323 patent was filed, AD researchers had largely abandoned tramiprosate to focus on more promising, unrelated, compounds. Opp. Exclude 4 (citing Ex. 2005, 249–50; Ex. 2007, 128; Ex. 2002, 9). According to Patent Owner, the Petition's failure to evaluate the state of the art relevant to what compounds a POSA would have considered most promising for treating AD in 2017 did not preclude Dr. Kimball from testifying about those critical facts. *Id.*

Patent Owner asserts that Dr. Kimball's testimony in this respect does not articulate "an unduly high standard that is inconsistent with the law." Opp. Exclude 5 (quoting Mot. Exclude, 6). Patent Owner contends that, "[f]irst, Petitioner fails to cite a single case supporting its argument that the above facts are inconsistent with a proper lead compound analysis." *Id.* (citing Mot. Exclude, 6). Second, argues Patent Owner, "Dr. Kimball confirmed he is not an attorney and was not trying to apply the 'legal lead compound test,' rather, he testified as to whether "whether a POSA would consider these molecules as leads." *Id.* (quoting Ex. 1069, p. 35, ll. 10–13). Patent Owner contends that the Board can simply apply the law to Dr. Kimball's opinion concerning the facts. *Id.*

We do not find Petitioner's arguments with respect to this point persuasive. Dr. Kimball is not an attorney qualified to practice before the Board or to render an opinion as to the correct application of the lead compound analysis. In other words, a lead compound analysis is a legal analysis, and one that it is the Board's to determine. Dr. Kimball's testimony as to whether a person of ordinary skill in the art would have selected tramiprosate or Val-APS for further development is an opinion that may be supported by the evidence of record. But the correct analysis to be performed, and its application to the facts of this case, are questions of law that are within the proper purview of the Board, which will weigh Dr. Kimball's opinions, and interpretation of the evidence, and then perform the requisite legal analysis. *See Science Applications Int'l Corp. v. United States*, 154 Fed.Cl. 594, 633 (Fed. Cl. 2021) (holding that

> [a]n expert may articulate the meaning of a term to a POSITA, but then the court must conduct a legal analysis to see if that same meaning fits with the term "in the context of the specific patent claim under review," because "experts may be examined to explain terms of art ... but they cannot be used to prove the proper or legal construction of any instrument of writing."

quoting *Teva Pharms. USA, Inc. v. Sandoz, Inc.*, 574 U.S. 318, 331 (2015) (internal quotations and citations omitted)).

At Dr. Kimball's deposition, the following colloquy occurred:

> Q. So am I correct in saying that you did not perform this step 2 analysis in your declaration?

> A. I am not a patent attorney. I didn’t apply a lead -- a legal lead compound test. I merely opined on the -- on whether a POSA would consider these molecules as leads.
>
> Q. Did you consider the question of a motivation to modify a lead compound in any way?
>
> A. Not in terms of a legal framework, no. I'm not -- again, I'm not a patent lawyer. I'm not applying a legal test.

Ex. 1069, p. 35, ll. 8–18. It is evident from this exchange, and related exchanges in the deposition transcript, that Dr. Kimball is permissibly opining as to the understanding of a person of ordinary skill in the art and not performing a legal analysis. We consequently do not find Petitioner’s arguments in this respect persuasive.

2. Dr. Kimball’s alleged failure to consider the prior art

Petitioner next argues that Dr. Kimball improperly ignored evidence of the state of the art as of March 2017. Mot. Exclude 7. Specifically, Petitioner alleges that Dr. Kimball excluded from consideration any evidence of the state of the art that that was not published before the filing date. *Id.* In support of this contention, Petitioner points to the following exchange from Dr. Kimball’s deposition:

> Q. Did you consider any studies other than those that were published prior to March 2017?
>
> A. I’ve only considered studies published prior to March 2017.

Id. (quoting Ex, 1069, p. 45, ll. 11–14).

Petitioner contends that Dr. Kimball cannot reliably opine whether a POSA would have selected tramiprosate, Val-APS, or compound 6 of Czarnik as a lead compound when he has failed to consider evidence of the state of the art as of the filing date that was published after the filing date. Mot. Exclude 7. Petitioner notes that a series of articles show that there was continued, ongoing interest in tramiprosate and Val-APS as of March 2017. *Id.* (*see* Reply 13–14).

Petitioner also argues that Dr. Kimball admitted that he was aware of, but excluded from his analysis, the disclosure of the ’323 patent that tramiprosate was "a promising investigational product candidate for the treatment of Alzheimer's disease" and stating that "it's only relevant what was available to a POSA." Mot. Exclude 8 (quoting Ex. 1069, pp. 106–109, ll. 18–9 (quoting Ex. 1001, col. 1, ll. 47–50)). Petitioner argues that "party admissions … are permissible evidence in an inter partes review for establishing the background knowledge possessed by a person of ordinary skill in the art." *Id.* (quoting *Qualcomm Inc. v. Apple Inc.*, 24 F.4th 1367, 1376 (Fed. Cir. 2022)). Petitioner alleges that Dr. Kimball applied improper methodology for understanding a skilled artisan's view of tramiprosate, Val-APS, and Czarnik's deuterium-labeled tramiprosate that renders his testimony about lead compound status unreliable. *Id.* at 9.

Patent Owner responds that, despite Petitioner's burden to establish the state of the art, none of the post-2017 evidence, or any of the evidence Dr. Kimball discussed about different AD compounds, was addressed in the Petition or its accompanying declaration. Opp. Exclude 5. Patent Owner argues further that Petitioner cites no authority supporting its assertion that evidence unavailable to a person of ordinary skill in the art at the time of invention is somehow relevant to determining which compounds a skilled artisan might have considered most promising in 2017. *Id.* at 6. Patent Owner asserts that determining whether a compound is a lead compound is evaluated as of the effective filing date of the patent. *Id.* (citing, e.g., *UCB v. Accord Healthcare*, 890 F.3d 1313, 1329 (Fed. Cir. 2018); *Pfizer v. Teva Pharms. USA*, 555 F. App'x 961, 971 (Fed. Cir. 2014)).

Patent Owner disputes Petitioner's contention that "Dr. Kimball also admitted that he was aware of, but excluded from his analysis, Risen's statement in the '323 patent that tramiprosate was 'a promising investigational product candidate for the treatment of Alzheimer's disease.'" Opp. Exclude 6 (quoting Ex. 1069, pp. 106–109, ll. 18–9 (quoting Ex. 1001, col. 1, ll. 47–50). Patent Owner acknowledges that the inventors believed the compounds to be promising or they would not have arrived at the invention. *Id.* at 7. Nevertheless, argues Patent Owner, the inventors' consideration is not the threshold test for the lead compound analysis. *Id.* (citing, e.g., *Otsuka Pharm. v. Sandoz*, 678 F.3d 1280, 1296 (Fed. Cir. 2012)).

We do not find Petitioner's arguments in this respect persuasive. During Dr. Kimball's deposition testimony, the following exchange took place:

> Q. And we also looked at the Bossu 2018 reference, Exhibit 1053; Sabbagh 2017, that was Exhibit 1055; and Martorana 2018, that was Exhibit 1056. You didn't consider any of those references in connection with preparing your declaration, did you?
>
> A. No, I did not. And to – to that point, they're not all – would not all have been available to a POSA, in any event.

Q. Isn't the work of researchers in this field relevant to understanding the state of the art?

A. Absolutely. And a POSA would know the state of – would have paid attention to the state of the art.… You have to take pretty seriously a large clinical trial that's run with over 1,000 people.

Q. So you're discounting anything published after March 2017; right?

….

A. It's not relevant to the position a POSA was in in March 2017. It was not knowledge anyone had.

Q. So is it correct to say that you've discounted that work?

….

A. I think it's correct – it's correct to say that I didn't see that. As of March 2017, I wouldn't have seen that work.

Q. So you haven't considered anything after March 2017 in connection with forming your opinion as to the state of the art at that time; is that correct?

A. Prior to this moment when you presented me with data that was published after 2017, the answer would be no.

Ex. 1069, pp. 103–105, ll. 17–13. It is evident from this exchange that Dr. Kimball was testifying as to what would have been the understanding of a person of ordinary skill in the art at the time of the filing of the '323 patent, in view of the art cited in the Wagner/Kimball Declaration, and did not consider the art published subsequent to the filing date, which would not have been within the knowledge of a skilled artisan at that time. We discern nothing objectionable in the testimony of Dr. Kimball that would warrant exclusion of the Declaration in this respect.

We come to the same conclusion with respect to the statement in the ’323 patent that “3-amino-1-propanesulfonic acid (also known as 3-APS, Tramiprosate, and Alzhemed™) is a promising investigational product candidate for the treatment of Alzheimer’s disease,” as discussed by Dr. Kimball. Ex. 1069, p. 106, ll. 21–24 (quoting Ex. 1001, col. 1, ll. 47–50). Dr. Kimball testified that he did not consider this statement because:

> A. Well, this would not have been available to a POSA in March because this would not have published until September, I would assume, based on the data -- the dates. So no one would have known of this. And, in any event, it’s the opinion of -- of a group that has a vested interest in the success of this.

Id. at p. 107, ll. 8–14. Dr. Kimball emphasizes that:

> A. My remit in the declaration -- I think it’s stated in -- if you go to paragraph 36 of my declaration -- is to provide an opinion on whether a POSA would view this as a lead candidate as of March 2017. So it’s only relevant what was available to a POSA.

Id. at p. 109, ll. 4–9. Again, we find nothing objectionable in Dr. Kimball’s testimony that would warrant exclusion of the Declaration.

3. Alleged Cursory Review and Possible Bias

Finally, Petitioner argues that the Kimball Declaration should be excluded as unreliable because Dr. Kimball disclosed at deposition that his review of the declaration was cursory at best, and thus insufficient to confirm the methodology used to reach the stated opinions, and because

Dr. Kimball's unexplained prior involvement with the case raises the possibility of bias. Mot. Exclude 9.

According to Petitioner, Dr. Kimball admitted at deposition that he spent a total of "three or four hours" on the declaration and that he adopted it without "hav[ing] firsthand knowledge of who actually wrote it" or doing anything to learn who did write it. Mot. Exclude 9 (quoting Ex 1069, pp. 13–14, ll. 21–9, pp. 15–16, ll. 16–2). Petitioner also notes that Dr. Kimball admitted that he "consulted the references in the declaration" but did not review any other materials. *Id.* (quoting Ex. 1069, p. 13, ll. 14–20).

Petitioner argues further that Dr. Kimball's disclosure that he was previously involved in this case for about a year in an unexplained capacity that involved "giving scientific advice about questions" to counsel for Patent Owner raises the possibility of bias. Mot. Exclude 10 (citing Ex. 1069, pp. 10–11, ll. 20–11, pp. 99–100, ll. 25–6). Petitioner alleges that Dr. Kimball's involvement extends beyond the normal role of a testifying expert in an IPR proceeding and argues that it is unclear what that role involves and what the full scope of Dr. Kimball's work for Patent Owner actually was. *Id.*

We are not persuaded by Petitioner's arguments. We have explained *supra* the unusual circumstances by which Dr. Kimball "stepped into the shoes" of Patent Owner's prior Declarant, the late Dr. Wagner. Dr. Kimball had approximately three weeks to prepare for deposition following the Board's February 2, 2022 Order allowing Patent Owner to substitute Dr. Kimball's Declaration for that of Dr. Wagner and, under the conditions of that order, Dr. Kimball's Declaration adopted that of Dr. Wagner's in its totality, with the exception of the relevant biographical details. *See* Paper

27, 2–3. Given the necessity of adopting another expert's declaration in toto, and of preparing for deposition on an accelerated schedule, it is not surprising that Dr. Kimball was not as well-prepared as an expert who had participated in Patent Owner's case since the inception of this proceeding.

The Board is capable of weighing Dr. Kimball's testimony in view of these circumstances, and we discern no legal basis for excluding Dr. Kimball's deposition, particularly in view of the likely prejudice to Patent Owner by being deprived of the Declaration of its principal expert. *See In re American Academy of Science Tech Center*, 367 F.3d 1359, 1370 (Fed. Cir. 2004) (holding that "the Board is entitled to give such weight to declarations as it deems appropriate").

Nor do we attach much probative value to Petitioner's vague allegation of potential bias. *See* Pet. 9. With respect to Petitioner's allegation, the following exchange took place:

> Q. Did you provide technical information related to those compounds to attorneys for Risen?
>
> ….
>
> A. I'm trying to recall. It's been a little while. Not as to specific compounds, I don't believe. Just more general scientific expertise.
>
> Q. So your -- the information you provided did not relate to tramiprosate, Val-APS, or Czarnik compound 6; is that correct?
>
> A. I did not provide an opinion on those specific molecules. That was not my charge.
>
> Q. Did your charge relate to the lead compound analysis in any way?

A. Not by me. I don't know what the -- what Wolf Greenfield did with it.

Q. Did your charge relate to the suitability of any of those compounds for use as a pharmaceutical compound?

A. My -- my charge was related to giving scientific advice about questions, not about opining on suitability of any particular molecule as a drug.

Ex. 1069, pp. 99–100, ll. 2–16. It is evident from this exchange that Dr. Kimball had performed some technical consulting work for Patent Owner that does not appear to be related to either the compositions or issues of the present trial.[5] Petitioner's vague allegation of "possible bias" in this case is without evidentiary support and consequently not persuasive.

In view of our foregoing analysis, Petitioner's Motion to Exclude the Declaration of Dr. Kimball is DENIED.

B. Level of Ordinary Skill in the Art

Petitioner asserts that a person of ordinary skill in the art would typically have a master's degree or a Ph.D. in chemistry, biochemistry, pharmaceutics, pharmaceutical science, or a related discipline. Pet. 6–7; Ex. 1002, 13–14. Alternatively, argues Petitioner, a skilled artisan might

[5] Subsequent to this colloquy, counsel for Patent Owner, Mr. Roses, instructed Dr. Kimball not to answer further questions concerning the nature of his consulting work, citing attorney-client privilege and noting that the questions were outside the scope of discovery. Ex. 1069, p. 100, ll. 8–19.

have a lesser degree, but more experience, and may have collaborated with others with specific skills in the art. *Id.* at 7; Ex. 1002, 14.

Patent Owner contends that a person of ordinary skill in the art would be a medicinal chemist with a Ph.D. and at least one year of work experience. PO Resp. 7 (citing Ex. 2042 ¶ 27; Ex. 2068 ¶ 27). Alternatively, argues Patent Owner, a skilled artisan might have a bachelor's or master's degree, but more experience. *Id.* Patent Owner asserts that, in either case, a person of skill in the art would have experience in preparing and evaluating compounds as potential treatments for Aβ-related diseases[6], such as Alzheimer's Disease ("AD"). PO Resp. 7 (citing Ex. 2042 ¶ 27; Ex. 2068 ¶¶ 26–27; Ex. 1001, Abstr.).

Patent Owner argues that, contrary to Petitioner's proposed definition, the ordinarily-skilled artisan would be a medicinal chemist, rather than the proposed different types of chemists (e.g., a general chemist, a biochemist) or a pharmaceutical scientist. PO Resp. 7. Moreover, argues Patent Owner, the requirement that on of ordinary skill in the art have experience in preparing and evaluating compounds as potential treatments for Aβ-related diseases, such as AD is supported by the disclosures of the '323 Patent and the teachings of Kong and Tolar. *Id.* All of these references, argues Patent Owner, employ compounds as pharmaceuticals for treating Aβ-related diseases, including Alzheimer's, as well as the synthesis, identification, isolation, and purification of isotopically-modified compounds. *Id.*

[6] The gradual accumulation of the amyloid-β protein in brain regions serving memory and cognition is hypothesized to be a precipitant of the earliest symptoms of Alzheimer's disease. *See* Ex. 2001, 1060.

According to Petitioner, this would require a specialized expertise unlikely to be within the general level of ordinary skill. *Id.* (citing Ex. 1001, col. 1, ll. 15–19; Ex. 1005, col. 1, ll. 15–21; Ex. 1008, col. 1, ll. 48–51; Ex. 2042 ¶ 28; Ex. 2041, p. 27, ll. 2–14).

Petitioner replies that Patent Owner's proposed definition requiring a "medicinal chemist" falls within Petitioner's definition of a person of ordinary skill in the art. Pet. Reply 9. Petitioner points to the testimony of its expert, Dr. Guengerich, who opines that "applying Patent Owner's definition of POSA would not change any of my opinions regarding the obviousness of the claims." *Id.* (quoting Ex. 1046 ¶ 7; also citing Ex. 1069, p. 19, ll. 18–21 ("You learned the medicinal chemistry on the job.")).

Having weighed the evidence and arguments on both sides, we define a person of ordinary skill in the art as:

> A person possessing a Ph.D. in chemistry, biochemistry, pharmaceutics, pharmaceutical science, or a related discipline with at least one year's experience in preparing and evaluating compounds as potential treatments for Aβ-related diseases, such as AD. Alternatively, a skilled artisan might have a lesser degree in these fields, with correspondingly longer experience.

Patent Owner adduces no evidence that a "medicinal chemist" would have skills that others in the disciplines proposed by Petitioner would not have, and which would make "medicinal chemists" uniquely qualified to be persons of ordinary skill in the art. Indeed, Patent Owner's arguments are undermined by the testimony of its own expert, Dr. Kimball:

> Q. Do you have a degree in medicinal chemistry, Doctor Kimball?

A. My degree is in organic chemistry and chemical biology from SUNY at Stony Brook.

Q. Is that a degree in medicinal chemistry, Doctor Kimball?

A. The degree in -- there are very few degrees in medicinal chemistry anywhere, right. Medicinal chemists were hired consistently in pharma and in biotech as synthetic organic chemists. Synthetic organic chemistry was the primary prerequisite for becoming a medicinal chemist. You learned the medicinal chemistry on the job.
….
Medicinal chemists are generally considered to be people who have a PhD in organic chemistry and have worked in the industry.

Ex. 1069, pp. 19–20, ll. 8–5.

Nevertheless, we agree with Patent Owner that a person of ordinary skill in the art would possess at least a year's worth of experience in preparing and evaluating compounds as potential treatments for Aβ-related diseases, such as AD. Such experience would be necessary for a person with the requisite advanced degree to become familiar with the literature and technical issues relating to this particular field of research.

C. Claim Construction

We construe claim terms so as to give the words in a claim their plain meaning, which is defined as the meaning understood by a person of ordinary skill in the art after reading the entire patent. *See* Changes to the Claim Construction Standard for Interpreting Claims in Trial Proceedings Before the Patent Trial and Appeal Board, 83 Fed. Reg. 51,340, 51,358 (Oct. 11, 2018) (amending 37 C.F.R. § 42.100(b) effective November 13, 2018) (now codified at 37 C.F.R. § 42.100(b) (2019)); 37 C.F.R. § 42.100(b); *see also Phillips v. AWH Corp.*, 415 F.3d 1303, 1312–13 (Fed. Cir. 2005) (en banc).

Both parties agree that the claim terms of the ’323 patent should be given their plain meaning, as the meaning understood by a person of ordinary skill in the art after reading the entire patent. Pet. 7; PO Resp. 8. We consequently determine that no express construction of any claim term is necessary to resolve the issues in dispute.

D. Principles of Law

A claim is unpatentable under 35 U.S.C. § 103 if the differences between the claimed subject matter and the prior art are such that the subject matter, as a whole, would have been obvious at the time the invention was made to a person having ordinary skill in the art to which the subject matter pertains. *KSR Int’l Co. v. Teleflex Inc.*, 550 U.S. 398, 406 (2007). The question of obviousness is resolved on the basis of underlying factual determinations, including: (1) the scope and content of the prior art; (2) any differences between the claimed subject matter and the prior art; (3) the level of skill in the art; and (4) if in evidence, so-called secondary considerations. *Graham v. John Deere Co.*, 383 U.S. 1, 17–18 (1966).

“In an [*inter partes* review], the petitioner has the burden from the onset to show with particularity why the patent it challenges is unpatentable.” *Harmonic Inc. v. Avid Tech., Inc.*, 815 F.3d 1356, 1363 (Fed. Cir. 2016). Although the burden of production may shift, the burden of persuasion on the issue of patentability remains with Petitioner and does not shift to Patent Owner. *Dynamic Drinkware, LLC v. Nat’l Graphics, Inc.*, 800 F.3d 1375, 1378 (Fed. Cir. 2015). To prevail, Petitioner must support its challenge by a preponderance of the evidence. 35 U.S.C. § 316(e); 37 C.F.R. § 42.1(d). We analyze the challenges presented in the Petition in accordance with these principles.

E. Grounds 1 and 2: Patentability of claims 1–17, 19, and 20 over Kong and Czarnik (Ground 1) and Patentability of claims 1–20 over Kong, Czarnik, and Tolar (Ground 2): Petitioner's arguments.

1. Overview of Kong

Kong issued on June 10, 2014 and is prior art to the '323 patent; indeed, the Specification of the '323 patent expressly incorporates Kong, in its entirety, by reference. *See* Ex. 1005; Ex. 1001, col. 2, ll. 8–11.

Kong is directed to methods, compounds, compositions and vehicles for delivering tramiprosate in a subject and, specifically, compounds that will yield or generate tramiprosate *in vitro* or *in vivo*, i.e., a prodrug. Ex. 1005, col. 1, ll. 15–20. Preferred compositions taught by Kong include amino acid prodrugs of tramiprosate. *Id.* at col. 1, ll. 20–22. Kong expressly teaches that among these preferred compositions is Val-APS, in which a valine residue is linked to the amino terminus of tramiprosate *via* a condensation reaction. *See id.* at col. 22, ll. 53–56, col. 24, ll. 15 Table 1 (compound A2), col. 27, ll. 58–61. Kong also demonstrates that administration of the tramiprosate prodrug Val-APS significantly increases the bioavailability (C_{max} and AUC) of tramiprosate, compared to administration of tramiprosate alone. *Id.* at col. 110, ll. 3–18, Table 10.

Kong further teaches that:

> The disclosed compounds also include isotopically labeled compounds where one or more atoms have an atomic mass

> different from the atomic mass most abundantly found in nature. Examples of isotopes that may be incorporated into the compounds of the present invention include, but are not limited to, $^{2}H(D)$, $^{3}H(T)$, ^{11}C, ^{13}C, ^{14}C, ^{15}N, ^{18}O, ^{17}O, etc.[7]

Ex. 1005, col. 7, ll. 46–52. Kong teaches that its compositions may specifically be used in the treatment of Alzheimer's disease. *Id.* at col. 68, ll. 5–30.

2. Overview of Czarnik

Czarnik is directed to deuterium-enriched tramiprosate and formed the sole basis of the Examiner's Final Rejection during prosecution. Ex. 1007 ¶¶ 2–3; Ex. 1004, 59. Czarnik is cited as a reference by the '323 patent. Ex. 1001, 1. Czarnik teaches embodiments of deuterated-substituted tramiprosate in which the hydrogens[8] of carbons 1–3 may be substituted with deuterium. Ex. 1007 ¶ 47, Table 2 (compound 6). Czarnik teaches that any or all of the hydrogens bound to carbons 1–3 may be substituted with deuterium: "Numerous modifications and variations of the present invention are possible in light of the above teachings." *Id.* ¶ 48; *see also* claims 1, 4. Specifically, Czarnik teaches:

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[7] We note that $^{2}H(D)$ is deuterium.

[8] Throughout the course of this Decision, we use the term "hydrogen(s)" to refer to the isotope of that element that is of greatest natural abundance, i.e., ^{1}H, having just a single proton as its nucleus. This isotope is also known as "protium." Deuterium or "heavy hydrogen," ^{2}H, possesses a proton and a neutron as its nucleus.

> The present invention is based on increasing the amount of deuterium present in tramiprosate above its natural abundance. This increasing is called enrichment or deuterium-enrichment. If not specifically noted, the percentage of enrichment refers to the percentage of deuterium present in the compound, mixture of compounds, or composition.… Since there are 9 hydrogens in tramiprosate, replacement of a single hydrogen atom with deuterium would result in a molecule with about 11 % deuterium enrichment.

Ex. 1007 ¶ 15. Czarnik also expressly discourages deuterium substitution at the amino and sulfonate terminal groups (i.e., R_1, R_2, and R_3 of tramiprosate):

> The hydrogens present on tramiprosate have different capacities for exchange with deuterium. Hydrogen atoms R_1-R_3 are easily exchangeable under physiological conditions and, if replaced by deuterium atoms, it is expected that they will readily exchange for protons after administration to a patient. The remaining hydrogen atoms are not easily exchangeable for deuterium atoms. However, deuterium atoms at the remaining positions may be incorporated by the use of deuterated starting materials or intermediates during the construction of tramiprosate.

Id. ¶ 14. Czarnik also expressly teaches that substitution with deuterium can occur at a single location on carbons 1–3 of the tramiprosate "backbone" (i.e., the carbons attached to R_4–R_9 in Czarnik's formula I):

> In another embodiment, the present invention provides a novel mixture of, deuterium enriched compound of formula I or a pharmaceutically acceptable salt thereof, wherein the abundance of deuterium in R_4-R_9 is at least 17%. The abundance can also be (a) at least 33%, (b) at least 50%, (c) at least 67%, (d) at least 83%, and (e) 100%.

Id. ¶ 32 (*see* also claim 4). Deuteration at a single site of R_4–R_9 would yield 17% deuteration.

Czarnik also teaches that deuterated tramiprosate can be used in the treatment of Alzheimer's disease. Specifically, Czarnik teaches that "[i]t is another object of the present invention to provide a method for treating Alzheimer's disease, comprising administering to a host in need of such treatment a therapeutically effective amount of at least one of the deuterium-enriched compounds of the present invention or a pharmaceutically acceptable salt thereof." Ex. 1007 ¶ 6.[9]

3. Overview of Tolar

Tolar is directed to "administration of certain compounds including Val-APS to ApoE4-positive patients for the treatment of Alzheimer's Disease." Ex. 1008, col. 1, ll. 65–67. Tolar recognizes that Val-APS is a prodrug of tramiprosate, and:

> [D]emonstrates key advantages over tramiprosate including improved GI tolerability, reduced nausea & vomiting, and decreased inter-subject PK variability. In addition, administering the prodrug provides for once-daily dosing due to an advantageously extended $t_{1/2}$ of 14.9 hours. Notably, oral administration of Val-APS provides a ~2-fold increase in plasma and brain exposure of tramiprosate in animals compared to tramiprosate administered at equimolar doses.

Id. at col. 1, ll. 56–64.

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[9] During prosecution, Patent Owner (then-Applicant) argued that "Czarnik […] is completely silent regarding the usefulness of deuteration of tramiprosate for therapeutic treatment, including of Alzheimer's disease." Ex. 1004, 53. We disagree for the reason stated in paragraph 6 of Czarnik and quoted above.

4. Ground 1: Patentability of claims 1–17, 19, and 20 over Kong and Czarnik

a. Introduction

In our Decision to Institute, we concluded that "Petitioner has established a reasonable likelihood of prevailing at trial" with respect to Ground 1. Dec. 19. We now consider Petitioner's arguments and supporting evidence in light of the additional arguments and evidence presented during the trial. *See* Pet. 26–50; PO Resp. 15–25.

Kong further teaches that tramiprosate is metabolized in humans, and other mammals to yield 2-carboxyethanesulfonic acid ("2-CES") and suggests that GABA transaminase and monoamine oxidase ("MAO") enzymes catalyze this reaction at the 3-carbon locus on the tramiprosate molecule, as indicated below:

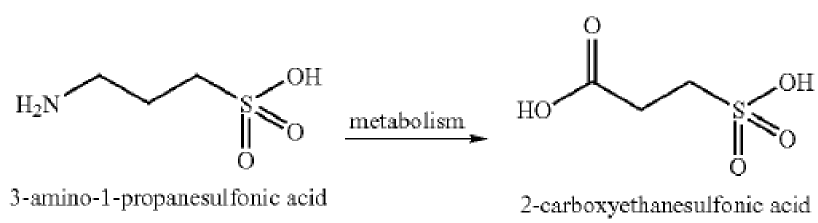


Figure illustrating metabolic degradation of tramiprosate to 2-CES, modified from Ex. 1005, col. 2, ll. 20–30

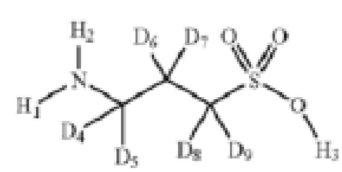
Kong also provides experimental verification of this method of degradation, demonstrating *in vitro* that degradation of tramiprosate to 2-CES was significantly inhibited by vigabatrin, a known GABA transaminase inhibitor, and also by nialamide, an inhibitor of monoamine oxidase. Ex. 1005, Example 5.

Furthermore, Kong teaches that the tramiprosate prodrug Val-APS (compound A2 in Kong) also significantly inhibits the metabolic degradation of tramiprosate. Ex. 1005, Example 3. Kong teaches that a two-fold increase in plasma concentration (C_{max}) and a significant increase in AUC of tramiprosate was observed when administered orally as Val-APS compared to tramiprosate. *Id.* at col. 110, 12–14, Table 10.

We find that Petitioner points to sufficient content within Kong as teaching both tramiprosate and a preferred prodrug, Val-APS, which fall within the scope of the chemical formulae II and I of independent claims 3 and 1, respectively, with the exception that, although Kong teaches isotope labeling (including deuterium) of tramiprosate and Val-APS, it does not expressly teach isotope labeling at the number 3 carbon position, as recited in the claims, i.e., "where R is a hydrogen of natural abundance, a deuterium (D) or a combination" (*see, e.g.*, Ex. 1001, col. 47, ll. 47–48). Pet. 26–29 (citing Ex. 1005, col. 24, l. 15, col. 1, ll. 50–51, col. 2, ll. 56–62, col. 68, ll. 5–30, col. 27, ll. 58–61, col. 22, ll. 53–56, col. 7, ll. 46–51).

Czarnik teaches deuterium-enriched tramiprosate for use in pharmaceutical compositions that may be used to treat Alzheimer's disease. Ex. 1007 ¶¶ 4–6, 8. The deuterium-substituted tramiprosate compositions of Czarnik includes substitutions at different positions, Czarnik's compound 6, which substitutes deuterium for natural-abundance hydrogen at each of the three carbons, is depicted below:

6



Compound 6 of Czarnik depicting deuterium (D) substitution at each of the three carbons of tramiprosate. Ex. 1007 ¶ 47

The reason for substituting deuterium for natural-abundance hydrogen is to maximize the kinetic isotope effect ("KIE"). As Petitioner's expert, Dr. Guengerich, explains:

> It has been known since the 1930s that replacing an atom in a chemical compound with a heavier isotope of the same element yields a stronger bond between that atom and the atom or atoms to which it is bonded.… Substituting a heavier isotope for the natural isotopic distribution is an established technique in the field for many years and is within the general knowledge of a POSA, as reflected in the literature.…
>
> Further, a POSA would recognize that substitution of deuterium for natural abundance hydrogen is expected to cause the most significant change because the relative difference in mass between the isotopes of hydrogen is greater than for any other element.

Ex. 1002 ¶¶ 47–48.

Dr. Guengerich explains that substitution of deuterium at a critical locus decreases the rate of metabolic degradation:

> [A] POSA would look to control the rate of metabolism to achieve various pharmacokinetic effects such as maintaining the active form longer, where the metabolite is inactivated, and reducing the rate of elimination, where the metabolite is more readily eliminated. Because the isotopically-substituted atom forms a stronger bond, as discussed above, the rate of metabolism and thus elimination of the compound from the subject will be reduced and the isotopically-enriched active will be expected to remain in the subject's system longer.

Ex. 1002 ¶ 57 (internal citation omitted).

In its response, Patent Owner argues Grounds 1 and 2 together. We consequently summarize Petitioner's arguments with respect to Grounds 1 and 2 before addressing Patent Owner's arguments in response.

b. Claim 1

With respect to claim 1 of the ’323 patent Petitioner argues that the only difference between Kong’s Val-APS and the ’323 Patent’s deuterated Val-APS, designated Compound 4 (i.e., D_2-Val-APS) and which is encompassed by claim 1, is substitution of deuterium for natural abundance hydrogen at the third carbon. Pet. 29. Petitioner asserts that it would have been obvious to a person of ordinary skill in the art to modify Kong’s Val-APS to substitute two deuterium atoms at the third carbon for two hydrogen atoms to form D_2-Val-APS for two reasons: (1) it is *prima facie* obvious to substitute one isotope of the same atom for another, and (2) the modification would have been obvious to a skilled artisan in view of Czarnik’s teaching of isotopically-substituted tramiprosate and Kong’s disclosure of positions at which isotopic substitution would be most effective. *Id.*

Concerning the first argument, Petitioner points to our reviewing court’s holding in *Takeda v. Alphaparm* that “structural similarity between claimed and prior art subject matter, proved by combining references or otherwise, where the prior art gives reason or motivation to make the claimed compositions, creates a prima facie case of obviousness.” *Takeda Chem. Indus., Ltd. v. Alphapharm Pty., Ltd.*, 492 F.3d 1350, 1356 (Fed. Cir. 2007). The *Takeda* court further held that “[a] known compound may suggest its homolog, analog, or isomer because such compounds ‘often have similar properties and therefore chemists of ordinary skill would ordinarily contemplate making them to try to obtain compounds with improved properties.’” *Takeda*, 492 F.3d at 1356 (quoting *In re Deuel*, 51 F.3d 1552, 1558 (Fed. Cir. 1995); *see also Anacor Pharm., Inc. v. Iancu*, 889 F.3d 1372, 1385 (Fed. Cir. 2018) (holding that “the greater the structural similarity between the compounds, the greater the motivation to combine and reasonable expectation of success"). We note here that the sum structural difference between the Val-APS of Kong and the D_2-Val-APS of claim 1 is precisely two neutrons. The same is true of tramiprosate and tramiprosate labeled with two deuterium atoms at the third carbon ("D_2-tramiprosate"). *See* EX1011, pp. 1-13–1-16.

With respect to its second obviousness argument, Petitioner contends that the modification of Kong's Val-APS to D_2-Val-APS would have been obvious to a person of ordinary skill in the art in view of Czarnik's teaching of isotopically-substituted tramiprosate and Kong's disclosure of positions at which isotopic substitution would be most effective. Pet. 30 (citing Ex. 1002 ¶¶ 89–91. As Dr. Guengerich explains:

> [A] POSA would recognize from Kong's teachings about the metabolism of tramiprosate that deuterium substitution should preferably be made at the third carbon (corresponding to the α-carbon in Val-APS) in order to slow metabolism of tramiprosate…. A POSA would anticipate that Val-APS would be metabolized to yield tramiprosate and tramiprosate would be metabolized…. Accordingly, a POSA would be likely to modify Kong's Val-APS prodrug according to Czarnik's disclosure of isotopically-substituted tramiprosate in compound 6 with deuterium substitution along the carbon chain to yield D_2-Val-APS.

Ex. 1002 ¶ 92.

Petitioner also argues that a person of ordinary skill in the art would have been motivated to modify Kong's Val-APS with isotopic substitution as taught by Czarnik to take advantage of the favorable KIE. Pet. 32; Ex. 1002 ¶ 93. Petitioner contends that it was well-known in the contemporaneous art that the enzymes known to metabolize tramiprosate exhibited a KIE in other reactions, and a POSA would specifically desire to slow metabolism of tramiprosate in view of Kong's teaching that extending exposure to tramiprosate provides an improved therapy. *Id.* (citing Ex. 1002 ¶ 68; Kong, col. 2, ll. 55–61).

Finally, Petitioner argues that a person of ordinary skill in the art would also have had a reasonable expectation of success in view of Kong's identification of locations for substitution and experimental evidence that tramiprosate is metabolized by enzymes known to exhibit significant KIE. Pet. 32; Ex. 1002 ¶ 94. Petitioner notes that Kong expressly contemplates isotopically-labelled compounds falling within the scope of its invention. *Id.* at 33 (citing Kong, col. 7, ll. 46–52). Petitioner argues that, because transaminase and monoamine oxidase are known to be affected by the KIE, a POSA would have a reasonable expectation of success in achieving the substitution and obtaining a slower rate of metabolism. *Id.*

c. Claims 2–17, 19, and 20

With respect to the claims 2–17, Petitioner argues that D_2-Val-APS is within the scope of claims 2, 4–6, 13, and 20 of the '323 patent. Pet. 33. Petitioner asserts that the claims are therefore obvious because a skilled artisan would have been motivated to modify Kong's Val-APS to form D_2-Val-APS, because the substitution is obvious in view of Czarnik's teaching of isotopic substitution in tramiprosate and Kong's teaching of the molecular site of the enzymatic metabolism at the number three carbon of tramiprosate by transaminases or oxidases, as described in the preceding section. *Id.*

Similarly, Petitioner contends, claims 3 and 19 of the '323 patent include within their scope tramiprosate, with isotopic substitution of nitrogen, oxygen and/or hydrogen at or near the site of enzymatic metabolism. Pet. 34. Petitioner asserts that these claims would have been obvious to a person of ordinary skill in the art in view of Kong's and Czarnik's teachings of tramiprosate, isotopic substitution of ^{15}N and deuterium at X and the third carbon, respectively, and the site of enzymatic metabolism, as well as the well-established knowledge in the art of the KIE as described in the preceding section. *Id.* Specifically, Petitioner argues that a person of skill in the art would have substituted ^{15}N for natural abundance nitrogen to yield a primary KIE and substituted ^{13}C, ^{17}O, and/or ^{18}O on the prodrug amide to yield a KIE associated with intermediate reaction steps or a secondary KIE. Pet. 35, 40, 42, 44.

Petitioner further asserts that claims 7–12 of the '323 patent include within their scope Val-APS with isotopic substitution of one or more of the nitrogen, carbon, or oxygen near the site of enzymatic metabolism. Pet. 33. Petitioner contends that the claims would have been obvious to a skilled artisan in view of Kong's and Czarnik's teachings of Val-APS, isotopic substitution, and the site of enzymatic metabolism, as well as of the kinetic isotope effect, as described in the previous section. *Id.*

Petitioner notes that claim 14 of the '323 patent depends from claim 1, and recites "level[s] of isotope enrichment in the compound with isotopes that are not of natural abundance" from about 2% to 100%. Pet. 45. Petitioner argues that Czarnik teaches percentages of substitution at a single site, and further teaches substitution between 11% and 100%, referring to total percentage deuterium among hydrogen atoms in the compound. *Id.* (citing Ex. 1007 ¶¶ 12, 15; *see also, e.g.*, claims 1–10, 14–16). Petitioner's declarant, Dr. Guengerich, opines that a person of ordinary skill in the art would appreciate that, if the substitution rate at the third carbon is low, the reaction will tend to proceed in the unsubstituted portion, severely limiting the significance of the kinetic isotope effect. *Id.* (citing Ex. 1002 ¶¶ 50–54). Petitioner therefore contends that a person of skill in the art would have been motivated to use higher percentages at the third carbon of isotopes that are not of natural abundance. *Id.*

Claim 15 of the ’323 patent recites “[a] pharmaceutical composition comprising the compound of claim 1 and a pharmaceutically acceptable carrier.” Pet. 46. Petitioner argues that this claim would have been obvious to a skilled artisan for the same reasons as claim 1, and Kong teaches that its compositions may be prepared “[i]n solid dosage forms … for oral administration” in which “the active ingredient is mixed with one or more pharmaceutically acceptable carrier….” *Id.* (citing Ex. 1005, col. 64, ll. 44–65, also citing Ex. 1005, col. 64, ll. 10–43; Ex. 1002 ¶¶ 124–125).

Claim 16 of the ’323 patent recites a composition of claim 15 that is suitable for oral administration and is in one of a list of specific dosage forms, including:

> [A] hard shell gelatin capsule, a soft shell gelatin capsule, a cachet, a pill, a tablet, a lozenge, a powder, a granule, a pellet, a pastille, a dragee, a solution, an aqueous liquid suspension, a non-aqueous liquid suspension, an oil-in-water liquid emulsion, a water-in-oil liquid emulsion, an elixir, or a syrup.

Ex. 1001, cols. 52–53, ll. 64–5.

Petitioner argues that claim 16 is obvious for the same reasons as claim 15, and because Kong teaches “[f]ormulations of [Kong’s] invention suitable for oral administration" that may be prepared in a variety of the same forms that are recited in claim 16. Pet. at 46–47 (citing Kong col. 64, ll. 26–33; also citing Ex. 1002 ¶¶ 126–127).

Claim 17 of the '323 patent is directed to a "method of treatment of an amyloid-β related disease" using a compound of claim 1 or the composition of claim 15. Petitioner argues that claim 17 would have been obvious to a skilled artisan because tramiprosate and prodrugs of tramiprosate, including Val-APS, were known in the art as treatments for amyloid-β related diseases such as Alzheimer's disease. *Id.* By way of example, Petitioner points to Kong's teaching that embodiments of its composition "pertain[ ] to a method for providing neuroprotection to a subject having an Aβ-amyloid related disease, e.g.[,] Alzheimer's disease." *Id.* (citing Ex. 1005, col. 68, ll. 5–10, col. 3, ll. 50–60).

5. Ground 2: Patentability of claims 1–20 over Kong, Czarnik, and Tolar

Petitioner essentially repeats the arguments presented in Ground 1 that claims 1–17, 19, and 20 would have been obvious over the combination of Kong and Czarnik. Pet. 52–53. Additionally, Petitioner argues, Tolar teaches that Val-APS is "an orally available small molecule prodrug of tramiprosate with improved pharmaceutical properties" that "demonstrates key advantages over tramiprosate including improved GI tolerability, reduced nausea & vomiting, and decreased intersubject PK variability." *Id.* at 52 (quoting Ex. 1008, col. 1, ll. 52–59). Petitioner argues that Tolar teaches that Val-APS provides an improved pharmacological profile and discloses clinical results in patients. *Id.* (citing Ex. 1008, col. 19, ll. 21–21, 24, Figs. 4–5, 7). According to Petitioner, Tolar teaches additional compounds with a single amino acid protecting group or two amino acids. *Id.* at 52–53 (citing Ex. 1008, col. 12–13, ll. 39–25, cols. 5–12, ll. 52–38, col. 4, ll. 8–10).

Claim 18 of the ’323 patent recites: The method of claim 17, “wherein the subject is ApoE4 positive.” Petitioner argues that claim 18 would have been obvious to a person of ordinary skill in the art over the combination of Tolar, Czarnik, and Kong, because Tolar teaches using Val-APS in “methods of treating and/or preventing Alzheimer’s disease in patients carrying one or two copies of the ApoE4 allele.” Pet. 59–60 (citing Ex. 1008, col. 5, ll. 26–29, also citing col. 18, ll. 19–27 (teaching preferably treating “ApoE4-positive patients”), claim 1; Ex. 1002 ¶¶ 176–177).

F. Patent Owner’s Response

1. Grounds 1 and 2

Patent Owner argues that Petitioner’s arguments, in Grounds 1 and 2, fail to meet its burden of proving unpatentability, because a proper analysis of obviousness under the lead compound test set forth in *Otsuka v. Sandoz* would not have led a person of ordinary skill in the art to select tramiprosate or Val-APS as a lead compound for further development. PO Resp. 1–2 (citing *Otsuka Pharm. v. Sandoz*, 678 F.3d 1280, 1291–93 (Fed. Cir. 2012)). Patent Owner points to the two-factor obviousness analysis set forth in *Otsuka* requires that a person of ordinary skill in the art:

> (1) select a prior art compound as a lead compound for further development; and

> (2) have a prior-art-based reason to modify that prior art compound in a manner that would produce a compound falling within the scope of the claim with a reasonable expectation of success.

Id. at 2 (citing *Otsuka*, 678 F.3d at 1291–93). Patent Owner notes that the second factor of the *Otsuka* analysis has two separate requirements: (1) a prior art-based (i.e., non-hindsight) reason to modify the prior art compound; and (2) a reasonable expectation of success in the asserted modifications. *Id.*

Patent Owner contends generally, with respect to all Grounds, that Petitioner's arguments fail for four reasons: (1) Petitioner fails to establish that a person of ordinary skill in the art would have selected tramiprosate or Val-APS (Grounds 1 or 2) or Czarnik compound 6 (Ground 3) as a "lead compound" for further development; (2) Petitioner fails to establish that a skilled artisan would have had a non-hindsight reason to modify tramiprosate or Val-APS or Czarnik compound 6 to arrive at the claimed compounds; (3) Petitioner fails to establish that a person skilled in the art would have had a reasonable expectation of successfully synthesizing, identifying, isolating, and purifying the claimed compounds, or of achieving a favorable kinetic isotope effect ("KIE") that would lead to improvements in pharmacological properties; and (4) there is objective evidence of failure of others, and unexpected results for the compounds the Petition alleges would have been obvious for a POSA to pursue, including D_2-tramiprosate and D_2-Val-APS. PO Resp. 2–5.

2. Tramiprosate and the lead compound analysis

Patent Owner argues that the *Otsuka* lead compound analysis is the correct one to apply, because the claimed compounds are "new chemical compounds." PO Resp. 11. Patent Owner contends that it is unaware of any case in which the Board or our reviewing court has found an isotopically-modified compound to be anticipated by its non-isotopically-modified analog. *Id.* at 12.

Patent Owner also argues that the chemical structures of the claimed compounds are not "unchanged" compared to tramiprosate and Val-APS and are therefore "new chemical compounds" to which the lead compound analysis should be applied. PO Resp. 12 (citing Ex. 2042 ¶ 54). Patent Owner contends that the properties of D_2-tramiprosate and D_2-Val-APS are different from their non-isotopically-modified forms. *Id.* (citing Petition, 32, 53, 62–63; Ex. 2041, p. 38, ll. 5–8, p. 43, ll. 14–17; also citing Ex. 1001, cols. 45–46, ll. 5–67). According to Patent Owner, if the properties of two compounds are different, the compounds themselves must therefore be different and their chemical structures cannot as a matter of law be "unchanged by the substitution of hydrogen isotopes." *Id.*

Patent Owner contends that all three of Petitioner's Grounds are fatally flawed in that they fail the first factor of the *Otsuka* analysis because the Petition fails to establish that a skilled artisan would have selected Val-APS or tramiprosate or Czarnik's compound 6 as "lead compounds" for further development. PO Resp. 14. Patent Owner maintains that a "lead compound" refers to "a compound in the prior art that would be most promising to modify in order to improve upon it … activity and obtain a compound with better activity." *Id.* (quoting *Takeda*, 492 F.3d at 1357. Patent Owner asserts that "mere structural similarity" between a prior art compound and a claimed compound is insufficient to establish the prior art compound as a "lead compound"—instead, the analysis is guided by evidence of the prior art compound's pertinent properties, including both positive attributes (e.g., activity, potency) and adverse effects (e.g., side effects, toxicity). *Id.* (citing *Otsuka*, 678 F.3d at 1292; *Daiichi Sankyo Co. v. Matrix Labs., Ltd.*, 619 F.3d 1346, 1354 (Fed. Cir. 2010)).

Patent Owner argues that Petitioner fails to prove that Val-APS, tramiprosate, or Czarnik compound 6 were "most promising" to modify to obtain a compound for treating AD. PO Resp. 14. Patent Owner argues that a person of ordinary skill in the art would have known, at the time of filing of the '323 patent, that: (1) tramiprosate had failed to meet its therapeutic endpoints in a Phase III clinical trial; (2) tramiprosate was shown to be unable to inhibit Aβ aggregation, the sole asserted mechanism of action for its purported biological activity, in even basic experiments excluding complex variables present in *in vivo* contexts; and (3) tramiprosate may promote tau aggregation and thus facilitate the pathology of AD. PO Resp. 14–15 (citing Ex. 2068 ¶ 66, 47–51, 68–70; Ex. 2034, 106–107; Ex. 2001, 1062; Ex. 2041, p. 59, ll. 16–19; Ex. 2004, 2, 9; Ex. 2075, 7; Ex. 2086, 7; Ex. 2087, 1951).

Patent Owner contends that, at the time of invention, multiple studies had demonstrated that tramiprosate failed to inhibit Aβ aggregation (the fundamental mechanism relied on for its purported activity), even when incubated with Aβ peptides *in vitro* and in the absence of metabolic enzymes. PO Resp. 15 (citing Ex. 2068 ¶¶ 50–51, 58–60, 81–82, 85; Ex. 2004, 2, 9; Ex. 2075, 7).

Patent Owner thus argues that a skilled artisan would not have selected tramiprosate as a lead compound because it was shown to promote tau aggregation, a negative effect that could facilitate the pathology of AD. *Id.* at 16 (citing Ex. 2068 ¶¶ 68–69; Ex. 2086, 7; Ex. 2087, 1951; also citing *Takeda*, 492 F.3d at 1359). Rather, Patent Owner argues, a person of skill in the art would have instead selected as a lead compound one of the compounds in Phase II or III clinical trials for AD or one with demonstrated mechanistic ability to modulate AD conditions. *Id.* (citing Ex. 2068 ¶¶ 91–94; Ex. 2004, 2 (bexarotene); Ex. 2075, 7 (epigallocatechin gallate); Ex. 2006, Abstract, Fig. 1; Ex. 2082 (compound 4); Ex. 2083 (BMS-986133 and BMS-932481); Ex. 2084 (LY2886721); Ex. 2085 (Verubecestat)).

Similarly, argues Patent Owner, a skilled artisan would not have selected Compound 6 of Czarnik as a lead compound because the record is devoid of any evidence suggesting that a person of skill in the art would have viewed Czarnik compound 6 to be promising or was ever tested in clinical trials. Pet. 17 (citing Ex. 2068 ¶ 88).

3. No non-hindsight-driven reason to modify

Patent Owner next challenges Petitioner's assertion that structural similarity alone is sufficient to establish a *prima facie* case of obviousness under the second factor of the lead compound analysis. PO Resp. 19–20 (citing Pet. 29). Specifically, Patent Owner disputes Petitioner's assertion that "structural similarity between claimed and prior art subject matter, proved by combining references or otherwise, where the prior art gives reason or motivation to make the claimed compositions, creates a prima facie case of obviousness." *Id.* at 20 (citing Pet. 29–30 (quoting *Takeda*, 492 F.3d at 1356)). Patent Owner contends that Petitioner quotes *Takeda* as holding that "[a] known compound may suggest its homolog, analog, or isomer," but omits the next sentence, which states that "to find a prima facie case of unpatentability … a showing that the 'prior art would have suggested making the specific molecular modifications necessary to achieve the claimed invention' [is] also required." *Id.* (citing Pet. 30 (quoting Takeda, 492 F.3d at 1356).

Patent Owner argues that Grounds 1–3 fail the second prong of the *Otsuka* lead compound analysis, which requires that the prior art have "supplied … a reason or motivation to modify a lead compound to make the claimed compound with a reasonable expectation of success." PO Resp. 21 (quoting *Otsuka*, 678 F.3d at 1292). Patent Owner argues that: (1) "[i]n addition to structural similarity between the compounds, a prima facie case of obviousness also requires a showing of 'adequate support in the prior art' for the change in structure;" and (2) Petitioner relies on a series of nine unsupported assumptions to arrive at the conclusion that a POSA would have been motivated to modify tramiprosate, Val-APS, or Czarnik compound 6 to obtain D_2-tramiprosate or D_2-Val-APS. *Id.* Each of these arguments is considered below.

First, Patent Owner argues that Kong's generic disclosure on isotopes provides no explanation of where on tramiprosate or its prodrugs a POSA should isotopically modify and no guidance as to which isotopes should be used. PO Resp. 24 (citing Ex. 2041, p. 125, ll. 10–21). According to Patent Owner, Kong's exemplary isotopes include not only unstable radioactive isotopes (e.g., ^{14}C) but also isotopes that Petitioner's expert acknowledged would be expected to lead to an inverse kinetic isotope effect that could worsen metabolic liabilities (e.g., ^{11}C). *Id.* (citing Ex. 2041, pp. 265–266, ll. 11–2). Furthermore, argues Patent Owner, despite disclosing a structurally undefined ^{14}C-3APS used in tracer studies, Kong does not disclose any specific isotopically-modified compounds or any deuterated compounds, or, specifically, one that is di-deuterated at the third carbon. *Id.* (citing Ex. 2041, p. 124, ll. 13–20, p. 125, ll. 14–17, p. 125, ll. 22–24).

Patent Owner argues further that Czarnik's similarly generic disclosure also fails to provide any guidance to a person of ordinary skill in the art because that reference discloses only two specific compounds as pharmaceuticals—hexa-deuterated Compound 6 and tramiprosate itself. PO Resp. 25 (citing Ex. 2068 ¶¶ 143–47; Ex. 2041, p. 147, ll. 12–18).

Patent Owner next contends that Petitioner relies on a string of allegedly unsupported, hindsight-driven assumptions to arrive at the conclusion that Kong, Czarnik, and Tolar would have taught a person of ordinary skill in the art to di-deuterate at tramiprosate or Val-APS at the third carbon, and to introduce other isotopic modifications to arrive at the challenged claims. PO Resp. 25. We summarize each of these assumptions alleged by Patent Owner below.

a. Tramiprosate would be metabolized by GABA-transaminase and/or methylamine oxidase ("MAO")

Patent Owner argues that Petitioner's statement that "Kong teaches that the 'metabolism of [tramiprosate] is mainly caused by a transaminase and/or monoamine oxidase [e.g., MAO]'" is incorrect, because Kong teaches that this is merely hypothetical. PO Resp. 27–28 (quoting Pet. 20–21, 32–33; Ex. 1002 ¶ 63).

Patent Owner criticizes the experimental evidence provided by Kong. Specifically, Patent Owner argues that Kon's *in vitro* data showing greater concentrations of 2-CES being produced from enzymatic degradation of tramiprosate than the concentration of the metabolite succinate produced by incubation of an equal concentration of γ-aminobutyric acid (“GABA”). PO Resp. 28 (citing Ex. 1005, col. 115, ll. 18–27. According to Patent Owner’s Declarant, Dr. Sessler, a person of ordinary skill in the art would question Kong’s hypothesis given the highly unusual result of an enzyme (purportedly GABA transaminase) metabolizing a synthetic substrate more rapidly, and to a greater extent, than its natural substrate. Ex. 2042 ¶¶ 78–79. Patent Owner also notes that Kong teaches that gabapentin, which increases GABA, had no effect on the conversion of tramiprosate to 2-CES. *Id.* (citing Ex. 1005, col. 115, ll. 33–36; Ex. 2042 ¶ 80–81). According to Dr. Sessler, a skilled artisan would have expected gabapentin to reduce the amount of tramiprosate metabolite present due to competition. *Id.* (citing Ex. 2042 ¶ 80.

Patent Owner also criticizes Kong’s disclosure that vigabatrin (a GABA transaminase inhibitor) and nialamide (an MAO inhibitor) reduce formation of 2-CES *in vitro*. Dr. Sessler attests that a person of ordinary skill in the art would have known that the allegedly neurotoxic effects of vigabatrin might affect the results, and would not understand of the latter experiment to suggest that MAO is involved in the enzymatic degradation of tramiprosate. PO Resp. 28–29 (citing Ex. 2042 ¶ 85). Patent Owner further notes that, because these studies of Kong were conducted in rat neuronal culture, other intermediate enzymes might have been involved in the formation of 2-CES. *Id.* at 29 (citing Ex. 1005, col. 115, ll. 18–36; Ex. 2042 ¶¶ 82–83, 89).

Petitioner asserts that, further undermining Kong, tramiprosate is not a "natural substrate" of either GABA transaminase or MAO, particularly given alleged the structural differences between tramiprosate and the natural substrates of GABA-T and MAO. PO Resp. 30 (citing Ex. 2042 ¶¶ 165–66). Petitioner argues that one of skill in the art would have appreciated that these differences could result in metabolism by different enzymes, which would prevent a skilled artisan from reasonably predicting whether a KIE would apply. *Id.* (citing Ex. 2042 ¶¶ 77, 94, 163, 207, 216–17; Ex, 2050, 108; Ex. 2059, 400).

b. Isotopic modification of tramiprosate would yield a KIE

Patent Owner next contends that Petitioner erroneously assumes that, because GABA-T and MAO have been found to exhibit KIEs with certain non-tramiprosate substrates, isotopic modification of tramiprosate would yield a KIE. PO Resp. 31 (citing Pet. 32–33).

Patent Owner argues that Petitioner cites no reference disclosing a KIE for metabolism of tramiprosate by GABA-T and/or MAO, but relies on references describing varying KIEs for various other substrates. PO Resp. 32 (citing Pet. 21–22 (citing Ex. 1024)). Dr. Sessler, Patent Owner's Declarant, argues that a person of ordinary skill in the art would have appreciated that these exemplary compounds bears little structural resemblance to tramiprosate, and a skilled artisan would not have reasonably expected any KIE-related findings with respect to that compound to apply to tramiprosate. *Id.* (citing Ex. 2042 ¶¶ 167–168).

Petitioner also argues that the direction and magnitude of a KIE obtained for a given enzyme-catalyzed reaction is complex and reliant on a number of factors, all of which are unknown for the interaction of tramiprosate and GABA transaminase or MAO. *Id.* at 33 (citing Ex. 2042 ¶¶ 182–183).

c. An expected KIE for tramiprosate would apply to tramiprosate prodrugs

Patent Owner next argues that Petitioner mistakenly assumes that any KIE obtained for tramiprosate would also apply to its prodrugs. PO Resp. 33. However, Patent Owner contends, the prodrug forms of tramiprosate are structurally different from tramiprosate itself, and, as a result, a skilled artisan would not reasonably expect to extrapolate any metabolic mechanism (and any associated KIE) from one to the other. *Id.* (citing Ex. 2042 ¶ 205).

Patent Owner points to the teachings of Kong, which demonstrate that different amino acid prodrugs of tramiprosate exhibit different drug bioavailabilities and blood stabilities, indicating that structurally distinct prodrugs of the same active are metabolized differently. PO Resp. 34 (citing Ex. 1005, col. 109, ll. 10–56 (Table 9), col. 108, ll. 1–44 (Table 8); Ex. 2042 ¶ 99). Patent Owner asserts that other prior art that metabolism of a compound by a particular enzyme may not translate to metabolism of structurally distinct compounds by the same enzyme. *Id.* (citing Ex. 2042 ¶ 101; Ex. 2051, 332, 343).

d. A KIE obtained for tramiprosate and its prodrugs would be significant and not inverse

Patent Owner asserts that Petitioner next assumes that a KIE obtained for tramiprosate and its prodrugs would be significant and not inverse. PO

Resp. 34. However, Dr. Sessler attests, the direction and magnitude of KIE are unpredictable, and depend on the identity of the enzyme, the substrate, and the kinetics of the enzymatic reaction. *Id.* at 34–35 (citing Ex. 2042 ¶¶ 96–97, 169–175, 178–183; Ex. 2041, p. 212, ll. 11–14, col. 214, ll. 5–8; Ex. 2058, 19502; Ex. 2057, 2875; Ex. 2038, 595). Petitioner contends that, Given the unpredictability of KIE effects, a person of ordinary skill in the art would not assume that isotopic modification of a tramiprosate or its prodrugs would result in a significant, favorable KIE. *Id.* at 35.

e. Other steps required to convert tramiprosate and its prodrugs to 2-CES would not impact the desired KIE

Patent Owner argues that Petitioner next assumes that other steps required to convert tramiprosate and its prodrugs to 2-CES metabolite observed in Kong would not impact the desired KIE. PO Resp. 35.

Patent Owner repeats Dr. Sessler's opinion that a person of ordinary skill in the art would have recognized other enzymes than GABA transaminase and MAO would necessarily be involved in the enzymatic catabolism of tramiprosate to 2-CES. PO Resp. 35 (citing Ex. 2042 ¶ 83). Patent Owner argues that a skilled artisan would know that the three-step process proposed by Petitioner's Declarant, Dr. Guengerich, would be one out of numerous possibilities, each step of which could be carried out by different enzymes catalyzing different reactions. *Id.* (citing Ex. 2042 ¶¶ 83, 109–110, 188). A skilled artisan, Patent Owner's Declarant opines, could not determine from Petitioner's expert's scheme which step is rate-limiting and potentially subject to a KIE without information on the kinetics of each reaction. *Id.* at 36 (citing Ex. ¶¶ 177, 207).

f. A KIE for tramiprosate and its prodrugs would provide a beneficial effect *in vivo*

Patent Owner next argues that Petitioner next erroneously assumes that a KIE for tramiprosate and its prodrugs would provide a beneficial effect *in vivo*. PO Resp. 36.

Patent Owner contends that a person of ordinary skill in the art would have appreciated that a KIE predicted based solely on the structure of a compound cannot necessarily be expected to result in a KIE *in vitro* or *in vivo*. PO Resp. 36 (citing Ex. 2042 ¶ 116; Ex. 1021, 379). According to Patent Owner, this is because predicted KIEs and even those providing *in vitro* changes to the performance of a compound may not necessarily translate into complex *in vivo* systems. *Id.* (citing Ex. 2042 ¶ 117, 158–61; Ex. 1029, 440; Ex. 2041, p. 92, ll. 21–23, pp. 225–226, ll. 11–1). Accordingly, Patent Owner asserts, a person of skill in the art would not have been able to assume that isotopic modification of tramiprosate or its prodrugs would provide a beneficial effect *in vivo*. *Id.*

g. Isotopic modification would not impact how, where in the body, and to what extent the prodrug is cleaved

Patent Owner challenges Petitioner's alleged assumption that isotopic modification of a tramiprosate prodrug would not impact how, where in the body, and to what extent the prodrug is cleaved. PO Resp. 37.

Dr. Sessler, Patent Owner's Declarant, opines that a person or ordinary skill in the art would have understood that isotopic modification of a prodrug can result in changes in residence times in different locations relative to the non-modified form. PO Resp. 37 (citing Ex. 2042 ¶ 114; Ex. 2041, p. 94, ll. 17–20, p. 96, ll. 5–10, p. 172, ll. 9–14). Consequently, Patent Owner argues, a skilled artisan would not have been able to predict how isotopic modification might influence tramiprosate prodrug cleavage and absorption of the active drug. *Id.* (citing Ex. 2042 ¶¶ 105–108, 115, 209, 245–46).

In support of this argument, Patent Owner points to the Specification of the '323 patent, which, argues Patent Owner, demonstrates that inclusion of ^{18}O at the carbonyl of the protecting group can produce deleterious effects for prodrugs of tramiprosate. PO Resp. 38 (citing Ex. 1001, Tale 8). According to Patent Owner, Table 8 demonstrates that ^{18}O substitution of the phenylalanine prodrug of tramiprosate reduces prodrug cleavage dramatically relative to di-deuteration at the third carbon, with a concomitant reduction in drug exposure. *Id.* Patent Owner argues that these data would confirm a skilled artisan's concern that isotopic modification (such as substitution of heavy isotopes on a prodrug moiety) might slow drug release, which is critical to obtaining the desired pharmacological activity. *Id.* Patent Owner argues that a person of skill in the art would have understood that these results are consistent with the understanding in the art of the unpredictability of isotopic substitution. *Id.* (citing Ex. 2042 ¶¶ 245–247).

h. A person of ordinary skill in the art would have only replaced two hydrogen atoms at C3 with deuterium atoms to the exclusion of other modifications

Patent Owner argues Petitioner erroneously assumes that a skilled artisan would have only di-deuterated at the third carbon of tramiprosate

("C3"), and would not have instead monodeuterated at C3 or isotopically modified other atoms, such as C3 itself or the hydrogen atoms bonded to C1 or C2. PO Resp. 39 (citing Pet. 31–32).

Patent Owner argues that Petitioner fails to explain why a person of skill in the art would have been motivated to modulate tramiprosate metabolism by isotopic modification would not have: (1) mono-deuterated at C3; (2) substituted ^{13}C at C3; and (3) deuterated at C2 and/or C1. PO Resp. 39 (citing Ex. 2042 ¶¶ 210, 231–233). With respect to (1), Dr. Sessler, notes that it was known in the art at the time of invention that certain enzymes have pro-chiral substrate selectivity, for which only a single deuterium would be needed to obtain a KIE. *Id.* (citing Ex. 2042 ¶ 232; Ex. 1022, 1; Ex. 2065, 14238; Ex. 1028, 1035).

With respect to (2), Patent Owner contends that Petitioner has taken the position that certain enzymes have pro-chiral substrate selectivity, for which only a single deuterium would be needed to obtain a KIE. *Id.* (citing Ex. 2042 ¶ 232; Ex. 1022, 1; Ex. 2065, 14238; Ex. 1028, 1035). And with respect to (3), Patent Owner argues that it is undisputed that a skilled artisan would have considered introducing deuterium at C2 in view of Petitioner's express position that "[a] POSA looking to gain the maximum isotope effect would [ ] be motivated to substitute heavier isotopes at other positions near [C3] in order to take advantage of the secondary KIE." *Id.* at 40 (citing Ex. 2042 ¶ 231; Pet. 16, 17; Ex. 2041, pp. 168-169, ll. 20–2, p. 176, ll. 10–17, 19–22).

Patent Owner contends that these modifications are consistent with the teachings of Kong and Czarnik, but would have provided compositions outside the scope of the challenged claims. PO Resp. 40 (citing Ex. 2042

¶¶ 210, 128; Ex. 2041, p. 125, ll. 18–21, pp. 148–149, ll. 7–15). Patent Owner notes that Petitioner's Declarant, Dr. Guengerich, opined that he would have expected Czarnik compound 6—which includes deuterium atoms at every position of C1, C2, and C3—to perform the same or better as a tramiprosate analog di-deuterated only at C3. *Id.* (citing Ex. 1007 ¶ 47; Ex. 2041, pp. 174–176, ll. 22–17, p. 180, ll. 2–6.

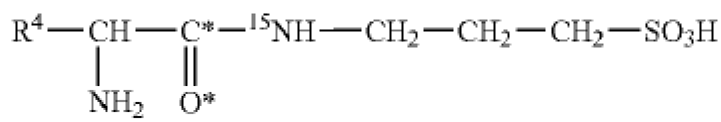
i. Di-deuteration at C3 would have provided an advantageous pharmacological effect *in vivo* without the addition of a prodrug

Finally, Patent Owner contends that Petitioner assumes that, for claims 3 and 19, di-deuteration at C3 would have provided an advantageous pharmacological effect *in vivo*, even without the addition of a prodrug moiety. PO Resp. 41.

Patent Owner argues that this assumption would require a person of skill in the art to ignore the stated benefits of tramiprosate prodrugs, as taught by Kong and Tolar. PO Resp. 41 (citing Ex. 2042 ¶¶ 48, 259, 261). According to Patent Owner, Kong teaches that amino acid prodrugs of tramiprosate are preferred, stating that those compounds "were able to prevent first-pass metabolism" of tramiprosate. *Id.* (quoting Ex. 1005, col. 109, ll. 58–62, also citing col. 1, ll. 20–22, col. 2, ll. 55–61; Ex. 2041, p. 73, ll. 7–10, pp. 73–74, ll. 24–18). Patent Owner also points to Tolar's teaching that Val-APS decreased tramiprosate metabolite in humans as compared to tramiprosate. *Id.* (citing Ex. 1008, col. 1, l. 56). Patent Owner therefore argues that a skilled artisan would not have relied exclusively on isotopic modification, or discarded the disclosed benefits associated with tramiprosate prodrugs. *Id.* (citing Ex. 2041, p. 180, ll. 13–16, pp. 187–188, ll. 24–8).

4. Claims 9–12

Patent Owner next argues that Petitioner has failed to satisfy its burden to support a motivation to arrive at the subject matter recited in claims 9–12. PO Resp. 42. Claim 9 recites:

> 9. A compound of Formula (V), or a pharmaceutically acceptable salt or ester thereof
>
> 
>
> where:
>
> R^4 is a side chain of a natural or unnatural amino acid;
>
> 0* is an oxygen atom of natural abundance, an ^{18}O, an ^{17}O or a combination thereof; and
>
> C* is a carbon atom of natural abundance, a ^{13}C or a combination thereof;
>
> provided that the compound is not N-acetyl-3-amino-l-propanesulfonic acid.

Ex. 1001, cols. 48–49, ll. 66–13.

Patent Owner challenges Petitioner's position that claims 9–12 would have been obvious in view of Kong and Czarnik because a person of ordinary skill in the art would have been motivated to introduce isotopic substitution in the prodrug potion of Val-APS, to the exclusion of deuteration at C3. PO Resp. 42 (citing Pet. 42–44, 56–58). Patent Owner contends that this is at odds not only with Petitioner's argument that a POSA would be motivated to deuterate at C3, but also its statements that "[t]he isotope effect associated with substitutions of heavy isotopes for carbon, nitrogen, or oxygen is expected to be small, but may be about three percent" of that for deuterium. *Id.* at 42–43 (quoting Pet. 16).

Patent Owner notes that Petitioner's declarant, Dr. Guengerich, acknowledged that a skilled artisan's "primary interest would be at the C-3 position and those carbon hydrogen bonds putting on deuterium." *Id.* at 43 (citing Pet. 16, 35; Ex. 1002 ¶ 48; Ex. 2041, pp. 164–165, ll. 22–6, p. 180, ll. 13–16). Patent Owner therefore argues that Petitioner has failed to establish that a person of ordinary skill in the art would have been motivated to make the substitutions recited in claims 9–12.

5. Claims 2 and 7–12

Patent Owner also argues that Petitioner has also failed to demonstrate that a person of ordinary skill in the art would have been motivated to arrive at the compositions recited in claims 3 and 7–12. PO Resp. 43. Patent Owner contends that Petitioner's position, that a person of skill in the art would have substituted ^{15}N for natural abundance nitrogen to yield a primary KIE and substituted ^{13}C, ^{17}O, and/or ^{18}O on the prodrug amide to yield a KIE associated with intermediate reaction steps or a secondary KIE, is unsupported by the prior art. *Id.* (citing Pet. 35, 40, 42, 44).

Specifically, Patent Owner contends that Kong, Czarnik, or Tolar fail to disclose any specific compounds containing ^{13}C, ^{17}O, ^{18}O, or ^{15}N isotopes, and the only reference cited by Petitioner to support its position does not relate to GABA transaminase- or MAO-mediated enzymatic degradation. PO Resp. 43 (citing Ex. 2041, p. 125, ll. 14–24, p. 128, ll. 8–13, p.141, ll. 17–20; Ex. 2042 ¶ 239; Pet., 40 (citing Ex. 1030)). Patent Owner again argues Dr, Guengerich's acknowledgment that a skilled artisan would be primarily interested in introducing deuterium at C3. *Id.* at 44.

Patent Owner also challenges Petitioner's assertion that a person of skill in the art might have been motivated to prepare a 15N-containing compound for use as a tracer. PO Resp. 44. Patent Owner contends that Petitioner's argument relies on a reference that generically discloses using stable isotopes as tracers. *Id.* (citing Pet. 35–36 (citing Ex. 1018, 3–4)). Patent Owner argues that this reference, alone, is insufficient to establish motivation. *Id.*

G. Analysis of Grounds 1 and 2

1. Claims 1 and 3

Our obviousness analysis weighs the four factors set forth in *Graham v. John Deere Co.*, 383 U.S. 1 (1966): (1) the scope and content of the prior art; (2) the level of skill of a person of ordinary skill in the art; (3) the differences between the claimed invention and the teachings of the prior art; and (4) the extent of any objective indicia of non-obviousness. 383 U.S. at 17-18. We have defined factor (2) in Section II.B. We address Patent Owner's arguments with respect to factor (4) separately in Section III.J.

With respect to factors (1) and (3), and as we have explained, Kong teaches both tramiprosate (which falls within the scope of claim 3, and its preferred prodrug, Val-APS (which falls within the scope of claim 1), with the exceptions of the requisite isotopic substitution at the third carbon. *See*

Section III.E.1, *supra*. Kong also teaches that administration of the prodrug Val-APS results in greater bioavailability of tramiprosate, as demonstrated by increased C_{max} and AUC when compared to administration of tramiprosate. *Id.* Kong also provides experimental data supporting its suggestion that enzymatic degradation of tramiprosate may occur by the action of GABA transaminase or MAOs, and that, in mammals including humans, the principal metabolite of this degradation is 2-CES, formed by deamination and carboxylation at the number 3 carbon. Ex. 1005, col. 2, ll. 10–43, Example 3, Table 10.

Furthermore, Kong expressly teaches that its claimed compositions:

> [A]lso include isotopically labeled compounds where one or more atoms have an atomic mass different from the atomic mass most abundantly found in nature. Examples of isotopes that may be incorporated into the compounds of the present invention include, but are not limited to ^{2}H(D), ^{3}H(T), ^{18}O, ^{17}O, ^{11}C, ^{13}C, ^{14}C, ^{15}N, etc.

Ex. 1005, col. 7, ll. 46–51.

Czarnik teaches isotopic substitution at various loci on the tramiprosate molecule, including at the third carbon. *See* Section III.E.2 *supra*. Specifically, Czarnik encourages substitution of deuterium at the three carbons in the chain of the molecule (including the third carbon) and discourages substitution at the amino and sulfonic acid terminal functional groups, as these hydrogens are easily exchanged. *Id.* Czarnik teaches that substitution can be made at one, several, or all of the hydrogens along the three-carbon backbone of the tramiprosate molecule. *Id.*

In sum, we conclude that the combined teachings of Kong and Czarnik meet all of the limitations of the claims 1 and 3. Specifically, it is uncontested that the references teach the basic chemical structure of the tramiprosate and Val-APS molecules, as recited in the claims. Furthermore, the references expressly teach the requisite isotopic substitutions at the loci recited in the claims. Czarnik expressly teaches substitution of deuterium for natural abundance hydrogen at locations on the carbon backbone of the molecule, including at the CR_2 site recited in the claims. See Ex. 1007 ¶ 48. Kong also teaches substitution of isotopes in tramiprosate and Val-APS, including substitution of ^{15}N and ^{13}C and ^{18}O. *See* Kong, col. 4, ll. 46–51. We conclude that a person of skill in the art would have found it obvious to make those substitutions at loci close to the site at which Kong teaches degradative metabolism of tramiprosate or Val-APS occurs, *viz.*, adjacent to the third carbon. *Id.* at col. 2, ll. 39–54

We agree with Petitioner that a person of ordinary skill in the art, having comprehended these teachings of Kong and Czarnik, would have been motivated to make the deuterium-for-hydrogen substitution at the number three carbon of tramiprosate, or its prodrug Val-APS, because that is the site of enzymatic degradation of tramiprosate. As Petitioner's Declarant, Dr. Guengerich, points out:

> [A] POSA would recognize from Kong's teachings about the metabolism of tramiprosate that deuterium substitution should preferably be made at the third carbon (corresponding to the α-carbon in Val-APS) in order to slow metabolism of tramiprosate…. A POSA would anticipate that Val-APS would be metabolized to yield tramiprosate and tramiprosate would be metabolized…. Accordingly, a POSA would be likely to modify Kong's Val-APS prodrug according to Czarnik's disclosure of isotopically-substituted tramiprosate in compound 6 with deuterium substitution along the carbon chain to yield D_2-Val-APS.

Ex. 1002 ¶ 93.

We conclude that a person of ordinary skill in the art would have been motivated to combine the teachings of Kong and Czarnik to take advantage of the KIE, a phenomenon well known in the art, and to make the deuterium substitution at the third carbon of tramiprosate to help inhibit the enzymatic degradation, because Kong teaches that the third carbon is the most likely site of enzymatic attack.

We also conclude that a person of ordinary skill in the art would have had a reasonable expectation of success. Deuterium substitution, as taught by Czarnik for tramiprosate, was well known in the art, as explained by Dr. Guengerich. *See* Ex. 1002 ¶¶ 39–42. Indeed, the teachings of Schofield[10], one of Patent Owner's exhibits, would have reinforced the skilled artisan's reasonable expectation of success in this regard. Schofield teaches that:

> By contrast, the tryptase inhibitor AVE5638 13 was much more metabolically stable in vitro in its deuterated form than when unlabelled. These results indicate that it *could be of value to concentrate efforts in this area to molecules which are metabolised by a major pathway that involves enzymes of the amine oxidase family [i.e., MAO] or other low-capacity enzyme families*.

Ex. 2047, Abstr. (emphasis added). Relevantly, the metabolism of AVE5638 13 "is *dominated by oxidative deamination to the carboxylic acid* via aldehyde. In common with other compounds of this class and β tryptase

[10] J. Schofield et al., *Effect of Deuteration on Metabolism and Clearance of Nerispirdine (HP184) and AVE5638*, 23 BIOORGANIC & MED. CHEM. 3831–42 (2015) ("Schofield") Ex. 2047.

itself, *this transformation is mediated by semicarbazide sensitive amine oxidase* (SSAO)." *Id.* at 3832 (emphases added, internal references omitted). In other words, a member of the amine oxidase family of enzymes acts upon AVE5638 13 *via* the same molecular mechanism that monoamine oxidase (a large family of enzymes) is thought to employ in the conversion of tramiprosate to 2-CES (i.e., oxidative deamination to the carboxylic acid). Schofield expressly teaches that this enzymatic degradation is inhibited, and the biostability of the drug increased, when deuteration occurs at a carbon located adjacent to the site of enzymatic attack. *See id.* at Scheme 2.

Finally, Schofield teaches that:

> SSAO is a member of the amine oxidase family of enzymes which includes monoamine oxidases A and B and polyamine oxidase in addition to SSAO.29, 30 *The observation that site selective deuteration can have a marked effect on the metabolic stability of drug molecules metabolized by enzymes of this family is quite well known*.

Ex. 2047, 3839 (emphasis added).

We find that a person of ordinary skill in the art would understand from the teachings of Scofield that deuterating the carbon adjacent to the site of attack of MAO could reasonably be expected to increase the stability of tramiprosate. Similarly, we conclude that a person of skill in the art would have found that it would have been obvious and desirable to substitute ^{15}N for natural abundance nitrogen, because the nitrogen in tramiprosate and Val-APS is located adjacent to the site of metabolic degradation and a skilled artisan would be motivated to substitute the isotope to increase the KIE and minimize enzymatic degradation.

Tolar reinforces the teachings of Kong and Czarnik and also expressly teaches the administration of tramiprosate in patients carrying one or two copies of the ApoE4 allele, as recited in claim 18. *See* Ex. 1008, col. 5, ll. 27–29.

2. Claims 2–20

The deuterated tramiprosate prodrug, D_2-Val-APS, is within the scope of claims 2, 4–6, 13, and 20 of the '323 patent, and we find that these claims are obvious for the same reasons as claim 1 for Grounds 1 and 2.

Similarly, claims 3 and 19 include within their scope tramiprosate with isotopic substitution of nitrogen and/or hydrogen at or near the site of enzymatic metabolism. Petitioner asserts that these claims would have been obvious to a person of ordinary skill in the art in view of Kong's and Czarnik's teachings of tramiprosate, isotopic substitution, and the site of enzymatic metabolism, as well as the well-established knowledge in the art of the KIE, as described in the preceding section. Pet. 34. We find the evidence of record somewhat weaker with respect to these claims. Both Czarnik and Tolar are silent with respect to isotopic substitution of ^{15}N for natural abundance nitrogen. Kong's teaching with respect to isotopic substitution states:

> The disclosed compounds also include isotopically labeled compounds where one or more atoms have an atomic mass different from the atomic mass most abundantly found in nature. Examples of isotopes that may be incorporated into the compounds of the present invention include, but 50 are not limited to, $^{2}H(D)$, $^{3}H(T)$, ^{11}C, ^{13}C, ^{14}C, ^{15}N, ^{18}O, ^{17}O, etc.

Ex. 1005, col. 7, ll. 46–52. Nevertheless, Kong also teaches that the site of enzymatic attack is at the bond between the third carbon and the nitrogen of the amine group, as demonstrated in this diagram from Kong:

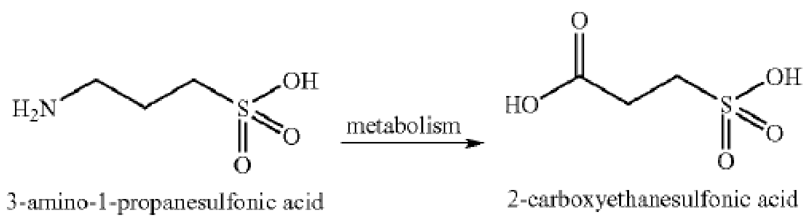


Diagram from Kong showing enzymatic degradation of tramiprosate to 2-CES. Other metabolites not found in mammals have been omitted

Id. at col. 2, ll. 20–30. We conclude that a person of ordinary skill would have understood from the teachings of Kong, as well as the general knowledge of isotopic substitution in the art, to substitute ^{15}N for N at the terminal amine group, as recited in claims 3. *See, e.g.*, Ex. 1017 ¶ 53; Ex. 2047, Abstr. Claim 1 recites substituting deuterium at the third carbon of tramiprosate. We conclude that the motivation to substitute the isotope in both of these claims, and the reasonable expectation of success, for a person of ordinary skill in the art would have been the same as the reasons that we have explained for claim 1 for Grounds 1 and 2.

Similarly, claims 7–12 of the ’323 patent include within their scope Val-APS with isotopic substitution of one or more of the nitrogen, carbon, or oxygen near the site of enzymatic metabolism. Pet. 33. For the same reasons we have explained for the preceding claims, we conclude that the claims would have been obvious to a skilled artisan in view of the prior art’s teachings of Val-APS as cited in Grounds 1 and 2, isotopic substitution, and the site of enzymatic metabolism, as well as of the kinetic isotope effect.

Dependent claim 14 of the ’323 patent recites “level[s] of isotope enrichment in the compound with isotopes that are not of natural abundance” from about 2% to 100%. Czarnik teaches percentages of substitution can be at a single site, and further teaches substitution between 11% and 100%, referring to total percentage deuterium among hydrogen atoms in the compound. *See* Ex. 1007 ¶¶ 12, 15, claims 1–10, 14–16. We conclude that, in view of these teachings of Czarnik, and for the reasons we have set forth *supra*, a person of ordinary skill in the art would have found challenged claim 14 to be obvious over the prior art cited in Grounds 1 and 2.

Claim 15 of the ’323 patent recites “[a] pharmaceutical composition comprising the compound of claim 1 and a pharmaceutically acceptable carrier.” Ex. 1001, col. 52, ll. 62–63. Kong teaches that its compositions may be prepared “[i]n solid dosage forms … for oral administration” in which “the active ingredient is mixed with one or more pharmaceutically acceptable carrier….” Ex. 1005, col. 64, ll. 44–65, col. 64, ll. 10–43. Czarnik also teaches “The present application describes deuterium-enriched tramiprosate, pharmaceutically acceptable salt forms thereof, and methods of treating using the same.” Ex. 1007, Abstr. Furthermore, and with respect to Ground 2, Tolar teaches administering tramiprosate to patients with AD, and a person of ordinary skill in the art would have understood that administering tramiprosate to a human patient would require the employment of a pharmaceutically acceptable carrier. *See, e.g.*, Ex. 1008, Abstr. Incorporating our reasoning set forth *supra*, we conclude that a skilled artisan would have found claim 15 obvious over the cited prior art in Grounds 1 and 2.

Claim 16 of the ’323 patent recites a composition of claim 15 that is suitable for oral administration and is in one of a list of specific dosage forms, including:

> [A] hard shell gelatin capsule, a soft shell gelatin capsule, a cachet, a pill, a tablet, a lozenge, a powder, a granule, a pellet, a pastille, a dragee, a solution, an aqueous liquid suspension, a non-aqueous liquid suspension, an oil-in-water liquid emulsion, a water-in-oil liquid emulsion, an elixir, or a syrup.

Ex. 1001, cols. 52–53, ll. 64–5

Kong teaches "[f]ormulations of [Kong's] invention suitable for oral administration" that may be prepared in a variety of the same forms that are recited in claim 16. *See* Kong, col. 64, ll. 26–34. We incorporate this teaching of Kong and our reasoning *supra* in concluding that claim 16 is obvious over the prior art cited in Grounds 1 and 2.

Claim 17 of the '323 patent is directed to a "method of treatment of an amyloid-β related disease" using a compound of claim 1 or the composition of claim 15. Ex. 1001, col. 53, ll. 6–20. Claim 18 (challenged in Ground 2) recites "[t]he method of claim 17, wherein the subject is ApoE4 positive." *Id.* at col. 53, ll. 21–22. It is undisputed that AD was known in the art at the time of filing to be an "amyloid-β related disease." For example, Tolar teaches that:

> Alzheimer's disease is characterized by two major pathologic observations in the brain: neurofibrillary tangles and beta amyloid (or neuritic) plaques, comprised predominantly of an aggregate of a peptide fragment know as beta amyloid peptide Aβ. Individuals with AD exhibit characteristic beta-amyloid deposits in the brain (beta amyloid plaques) and in cerebral blood vessels (beta amyloid angiopathy) as well as neurofibrillary tangles.

Ex. 1008, col. 1, ll. 35–42.

Kong teaches that embodiments of its composition "pertain[ ] to a method for providing neuroprotection to a subject having an Aβ-amyloid related disease, e.g.[,] Alzheimer's disease." Ex. 1005, col. 68, ll. 5–10, col. 3, ll. 50–60. Similarly, Czarnik teaches that "[i]t is another object of the present invention to provide a method for treating Alzheimer's disease, comprising administering to a host in need of such treatment a therapeutically effective amount of at least one of the deuterium-enriched compounds of the present invention or a pharmaceutically acceptable salt thereof." Ex. 1007 ¶ 6. Tolar also teaches administration of tramiprosate prodrug Val-APS to patients with Alzheimer's disease, as well as to "specific sub-populations that have one or more copies of the ApoE4 (or E4) allele" (an indicator for a genetic risk factor for patients with late-onset AD). Ex. 1008, col. 1, ll. 53–67, col. 5. ll. 5–8. Incorporating our reasoning presented *supra*, we conclude that a person of skill in the art would have concluded that claim 17 is obvious over the prior art cited in Grounds 1 and 2, and claim 18 is obvious over the prior art cited in Ground 2.

3. Deficiencies in Patent Owner's Arguments

Patent Owner advances several lines of argument against the obviousness of the challenged claims in Grounds 1 and 2, and we consider each of them in turn.

a. Tramiprosate as a lead compound

At the heart of Patent Owner's argument is the contention that deuterated tramiprosate and deuterated Val-APS would not have been obvious over the cited prior art, because a person of ordinary skill would not have selected either tramiprosate or its prodrug as a lead compound for development, because such a skilled artisan would have known that tramiprosate was not the most promising drug candidate for treating Alzheimer's Disease. *See* PO Resp. 14–16. Rather, Patent Owner argues, a person of skill in the art would have picked a more promising drug for development. *Id.*

Patent Owner argues that the Federal Circuit's holding in *Otsuka* sets forth the correct analytical framework for selecting only the most promising drug for development, and hence the only non-obvious choice. PO Resp. 8–9. As part of this argument, Patent Owner argues that D_2-Val-APS and D_2-tramiprosate are new compounds and not merely obvious variants of undeuterated Val-APS and tramiprosate, respectively. PO Resp. 11–12. Patent Owner states that it is aware of no case in which the Federal Circuit or the PTAB has found an isotopically-modified compound to be anticipated by its nonisotopically-modified analog. *Id.* at 12. Furthermore, Patent Owner contends, because a compound and its properties are inseparable, if the isotopically-substituted compositions manifest different properties than the undeuterated forms, they are per se different compositions.

We do not find these arguments persuasive. The question before us is not whether deuterated tramiprosate and Val-APS are anticipated by the undeuterated forms, but whether, in view of the teachings of the cited prior art, they are obvious over them. Although we are not aware of any decision by our reviewing court that squarely addresses the question of whether an isotopically-substituted compound is obvious over the non-substituted version, the Federal Circuit has held that "[a] known compound may suggest its homolog, analog, or isomer because such compounds 'often have similar properties and therefore chemists of ordinary skill would ordinarily contemplate making them to try to obtain compounds with improved properties.'" *Takeda*, 492 F.3d at 1356 (quoting *In re Deuel*, 51 F.3d 1552, 1558 (Fed. Cir. 1995)).

We think that this holding has considerable relevance in this instance. It is undisputed that deuterium is the same element as hydrogen (and tritium), these three isotopes all have the same atomic number and are not listed separately in any periodic table of the elements. The same is true for ^{14}C, ^{15}N, ^{18}O and the various isotopes of uranium, among many others. The sum of the chemical difference between D_2-tramiprosate/D_2-Val-APS and non-substituted tramiprosate/Val-APS is two additional subatomic particles (neutrons) in the former.

Indeed, it was generally known in the art at the time of invention that, when deuterium is incorporated into molecules in place of hydrogen, in most respects, the deuterated compound is very similar to the all-natural-abundance hydrogen compound. *See* L. Di Costanzo et al., *Expression, Purification, Assay, and Crystal Structure of Perdeuterated Human Arginase I*, 465(1) ARCH. BIOCHEM. BIOPHYS. 82 (2007) ("Di Costanzo"). According to Di Costanzo, the electron clouds of its component atoms define the shape of a molecule, deuterated compounds have shapes and sizes that are essentially indistinguishable from their all-natural-abundance hydrogen analog. And because deuterium is a ubiquitous and naturally-occurring element, it is not unlikely that a certain amount of deuterium exists in tramiprosate even when no experimental isotopic substitution is attempted. *See, e.g.*, Ex. 1007 ¶ 10 ("The natural abundance of deuterium is 0.015%. One of ordinary skill in the art recognizes that in all chemical compounds with a H atom, the H atom actually represents a mixture of H and D, with about 0.015% being D").

Moreover, both Kong and Czarnik teach deuterium substitution of tramiprosate and Val-APS with an intent of exploiting the KIE "to try to obtain compounds with improved properties." *Takeda*, 492 F.3d at 1356 (quoting *In re Deuel*, 51 F.3d 1552, 1558 (Fed. Cir. 1995)). We consequently conclude that the standard set forth in *In re Dillon* appears to be the closest applicable standard to apply in this case: *viz.*, that "structural similarity between claimed and prior art subject matter, proved by combining references or otherwise, where the prior art gives reason or motivation to make the claimed compositions, creates a prima facie case of obviousness." 919 F.2d 688, 692 (Fed Cir. 1990) (en banc). With respect to this standard, the Federal Circuit clarified in *Takeda* that "to find a prima facie case of unpatentability in such instances, a showing that the "prior art would have suggested making the specific molecular modifications necessary to achieve the claimed invention" was also required." 492 F.3d at 1356. In the present case, we conclude, for the reasons that we have explained, that deuteration of tramiprosate and Val-APS at the third carbon is a modification that Kong and Czarnik together directly suggest.

In *Otsuka*, the Federal Circuit set forth a two-part test for determining whether a skilled artisan would select a compound as a lead compound for development. First, in selecting a lead compound, the analysis is guided by evidence of the compound's pertinent properties. *Otsuka*, 678 F.3d at 1292. Secondly, the court looks to whether the prior art would have supplied one of ordinary skill in the art with a reason or motivation to modify a lead compound to make the claimed compound with a reasonable expectation of success. *Id.* (citing *Takeda*, 492 F.3d at 1357). In the present case, both tramiprosate and Val-APS were well known in the art, as taught by Kong, Czarnik, and Tolar, as promising compositions for the treatment of AD. And we have explained *supra* why a person of skill in the art would have been motivated to substitute deuterium for hydrogen at the third carbon, and why such an artisan would have had a reasonable expectation of success, *viz.*, to increase the KIE and to reduce tramiprosate's sensitivity to enzymatic degradation and hence increase its efficacy.

In contrast to the facts of the present case, the facts forming the basis of the *Otsuka* decision, upon which Patent Owner relies, are considerably different. In *Otsuka*, the compounds selected as "lead compounds" were selected form a vast genus encompassing "approximately nine trillion compounds." *Otsuka*, 678 F.3d at 1286. The Federal Circuit also found in *Otuska* that there lacked sufficient reasons for a person of skill in the art to select the contested compounds as a lead drug, finding that, for at least one, "[t]aken as a whole, however, the prior art taught away from using [the drug in question] as a starting point for further antipsychotic research." *Id.* at 1295.

In the present case, the facts are much different. Tramiprosate and its Val-APS prodrug were not plucked, seemingly at random, from an enormous genus of the prior art. Rather, Kong, Czarnik, and Tolar all expressly teach tramiprosate and/or Val-APS as a potential drug for treating AD, and Kong and Czarnik both suggest mechanisms by which the efficacy of tramiprosate may be increased, i.e., use of the prodrug and/or isotopic substitution to enhance the KIE. *See* Ex. 1005, col. 7, ll.41–52; Ex. 1007 ¶¶ 4–9; Ex. 1002, ¶¶ 39–42.

Patent Owner makes much of the fact that tramiprosate failed to meet the standard of success in its phase III clinical trials and argues that, therefore, a skilled artisan would not have selected it as a lead compound for further development. *See* PO Resp. 14–15. Patent Owner also points to several *in vitro* studies suggesting that tramiprosate may have failed to act in a manner predictive of success in preventing amyloid-β aggregation and promoted tau-aggregation. *Id.* at 15 (citing Ex. 2004, 2, 9; Ex. 2075, 7; Ex. 2001, 1062; Ex. 2086, 7; Ex. 2087, 1951).

Nevertheless, the evidence of record indicates that clinical interest in tramiprosate as an anti-AD drug remained substantial at the time of filing of the ’323 patent. As Petitioner points out, despite the alleged clinical trial failure of tramiprosate, one peer-reviewed article identified thirteen publications on clinical studies and subgroup analyses of tramiprosate between 2006 to 2018. Pet. Reply 13–14 (citing Ex. 1050[11], Table 2).

Tellingly, the Specification of the ’323 patent also discloses that “3-amino-1-propanesulfonic acid (also known as 3APS, Tramiprosate, and Alzhemed™) is a promising investigational product candidate for the treatment of Alzheimer’s disease. 3APS was the subject of Phase III clinical trials in North America and Europe.” Ex. 1001, col. 1., ll. 47–51. Patent Owner disputes that this statement constitutes an admission. *See* PO Resp.

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[11] S. Manzano et al., *A Review on Tramiprosate (Homotaurine) in Alzheimer’s Disease and Other Neurocognitive Disorders*, 11 FRONTIERS IN NEUROL., Article 614, 1–10 (July 2020) (“Manzano”) Ex. 1050.

6–7. Even so, we find that the statement exemplifies a person of skill in the art's knowledge of the state of the art at the time of filing. *See* Hearing Trans., p. 42, ll. 1–9. As the Federal Circuit held in *Qualcomm Inc. v. Apple Inc.*:

> We have held that "it is appropriate to rely on admissions in a patent's specification when assessing whether that patent's claims would have been obvious" in an *inter partes* review proceeding. *Koninklijke Philips N.V. v. Google LLC*, 948 F.3d 1330, 1339 (Fed. Cir. 2020) (citing *PharmaStem Therapeutics, Inc. v. ViaCell, Inc.*, 491 F.3d 1342, 1362 (Fed. Cir. 2007).
>
> That is because a petitioner may rely on evidence beyond prior art documents in an *inter partes* review, even if such evidence itself may not qualify as the "basis" for a ground set forth in a petition. *See* 35 U.S.C. § 312(a)(3)(B); § 314(a); § 316(a)(3); *Yeda Rsch. v. Mylan Pharms. Inc.*, 906 F.3d 1031, 1041 (Fed. Cir. 2018). Indeed, *Koninklijke Philips* specifically rejected an argument that the general knowledge of a skilled artisan may not be relied on in an *inter partes* review because it does not constitute "prior art consisting of patents or printed publications" under § 311(b). *See* 948 F.3d at 1337. In other words, the assessment of a claim's patentability is inextricably tied to a skilled artisan's knowledge and skill level.
>
> As a patentee's admissions about the scope and content of the prior art provide a factual foundation as to what a skilled artisan would have known at the time of invention, *Randal Manufacturing v. Rea*, 733 F.3d 1355, 1362–63 (Fed. Cir. 2013), it follows that [applicant admitted prior art, i.e., the specification of the challenged patent] may be used in similar ways in an *inter partes* review, *see McCoy v. Heal Sys., LLC*, 850 F. App'x 785, 789 (Fed. Cir. 2021).

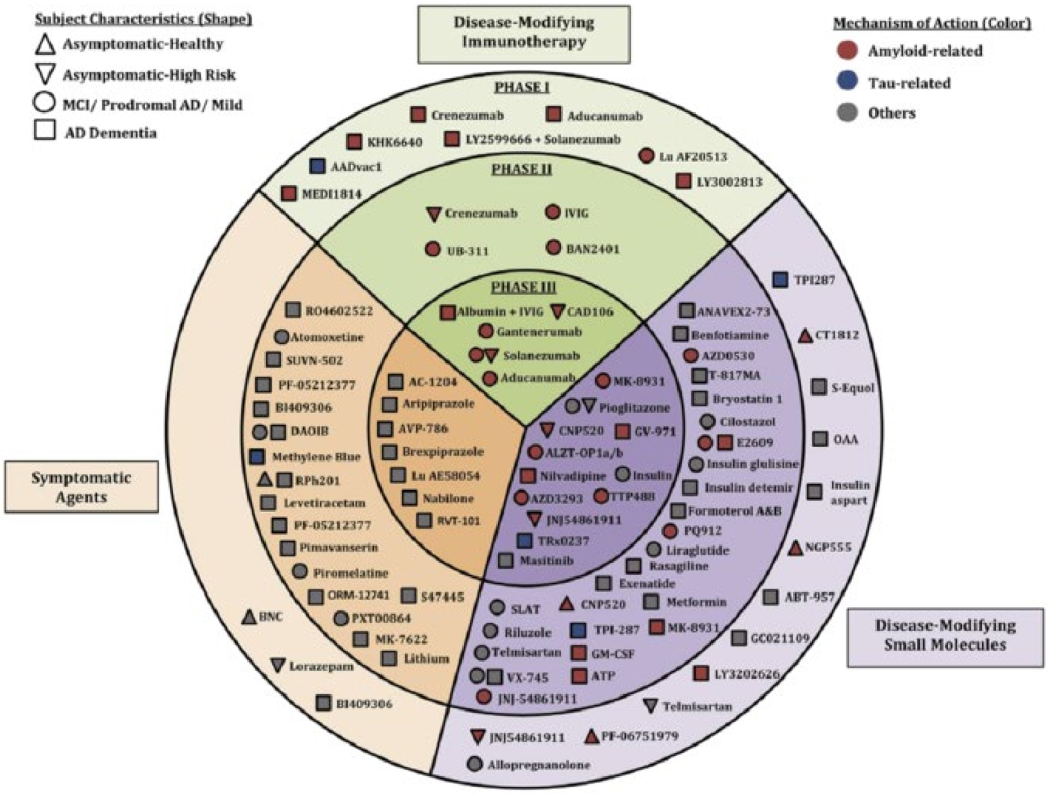
24 F.4th 1367, 1375–76 (Fed. Cir. 2022) (internal quotations and references omitted). We consequently find that we may properly consider the statements of the '323 patent regarding the potential promise of tramiprosate

as a drug for the treatment of AD insofar as it is related to the knowledge of a person of ordinary skill in the art at the time of filing. In particular, we consider the statements as they relate to the extent that a person of skill in the art would have understood that tramiprosate was a "a promising investigational product candidate for the treatment of Alzheimer's disease."

Furthermore, although tramiprosate may have failed to meet the criteria for success in its phase III clinical trials, no such evidence was adduced with respect to the prodrug Val-APS, which Kong teaches increases the C_{max} and AUC of circulating tramiprosate by conferring first-pass metabolic resistance on the drug. *See* Ex. 1005, Example 3, Table 10.

Finally Patent Owner's contention that only the most promising drug for the treatment of AD is the sole obvious choice for development is confounded by Patent Owner's own evidence. At oral argument, Patent Owner pointed to Figure 1 of Exhibit 2006[12], which is reproduced below:

[12] J. Cummings et al., *Alzheimer's Drug-Development Pipeline: 2016*, 2 ALZHEIMER'S & DEMENTIA: TRANSLATIONAL RES. & CLIN. INTERVEN. 222–32 (2016) ("Cummings") Ex. 2006.



<u>Figure 1 of Exhibit 2006 depicts agents in contemporaneous clinical trials for AD (shape indicates stage of disease of patients in the trials: color shows the mechanism of action: location shows phase of development and category of activity</u>

Simply put, how is the skilled artisan to choose from this plethora of different compounds, at different stages of development and with different modes of action, the sole compound, to the exclusion of the others, that "would be most promising to modify in order to improve upon its … activity and obtain a compound with better activity?" *Otsuka*, 678 F.3d at 1291 (quoting *Takeda*, 492 F.3d at 1357).

We conclude that, for the reasons we have explained *supra*, and given the teachings of the cited prior art, and the evidence of record concerning the knowledge of a person of ordinary skill in the art, a skilled artisan would have been motivated to make isotopic substitutions to tramiprosate and Val-APS to enhance the KIE and to enhance the efficacy of the drug and prodrug.

b. Grounds 1 and 2 and Patent Owner's nine assumptions

Patent Owner next argues that Kong, Czarnik, and Tolar fail to provide a reason to modify tramiprosate or Val-APD to arrive at the claimed compounds, contending that none of the references expressly teaches the specific D_2-tramiprosate and D_2-Val-APS compounds. PO Resp. 23–25. We have explained *supra* why we disagree with this argument, and why we find that the combined teachings of the references would make the compositions recited in the challenged claims obvious to a person of ordinary skill in the art. Patent Owner argues that Petitioner relies on nine erroneous assumptions in attempting to establish that the challenged claims are obvious over the prior art forming the basis of Grounds 1 and 2. We now turn to these arguments. We have summarized these arguments in Section III.F.3.a–i *supra*. To recapitulate, Patent Owner alleges that Petitioner wrongly assumes that:

(a) Tramiprosate would be metabolized by GABA transaminase and/or MAO;

(b) Isotopic modification of tramiprosate would yield a KIE;

(c) An expected KIE for tramiprosate would apply to tramiprosate prodrugs;

(d) A KIE obtained for tramiprosate and its prodrugs would be significant and not inverse;

(e) Other steps required to convert tramiprosate and its prodrugs to 2-CES would not impact the desired KIE;

(f) KIE for tramiprosate and its prodrugs would provide a beneficial effect *in vivo*;

(g) Isotopic modification would not impact how, where in the body, and to what extent the prodrug is cleaved;

(h) A person of ordinary skill in the art would have only replaced two hydrogen atoms at C3 with deuterium atoms to the exclusion of other modifications; and

(i) Di-deuteration at C3 would have provided an advantageous pharmacological effect *in vivo* without the addition of a prodrug.

See PO Resp. 27–41.

We have considered Patent Owner's arguments with respect to these alleged assumptions, but we do not find them persuasive. As an initial matter, we remind Patent Owner that "[o]nly a reasonable expectation of success, not absolute predictability, is necessary for a conclusion of obviousness." *In re Longi*, 795 F.2d 887, 897 (Fed. Cir. 1985). Indeed, "[w]hat does matter is whether the prior art gives direction as to what parameters are critical and which of many possible choices may be successful." *Allergan, Inc. v. Apotex Inc.*, 754 F.3d 952, 965 (Fed. Cir. 2014).

With respect to assumption (a), Kong provides not only suggestions, but experimental data that would reasonably suggest to a person of ordinary skill in the art that tramiprosate is vulnerable to enzymatic degradation at the third carbon *via* oxidative deamination to the carboxylic acid by MAO and/or GABA transaminase. *See* Ex. 1005, col. 2, ll. 8–54; Example 5.

Specifically, Kong teaches that, in *in vitro* hippocampal slice cultures, inhibitors of GABA transaminase (vigabatrin) and monoamine oxide (nialamide) reduced the formation of 2-CES. *Id.* at col 115, ll. 8–36. In contrast, gabapentin (known to increase GABA concentration in the brain) had no significant effect on the conversion of tramiprosate to 2-carboxyethane-sulfonic acid. *Id.* These data would suggest to a person of ordinary skill in the art that GABA transaminase and MAO catalyze the catabolism of tramiprosate to 2–CES. This is reinforced by the teachings of Schofield, one of Patent Owner's exhibits, that directly suggests that the oxidative deamination to carboxylic acid is mediated by amine oxidases. *See* Ex. 2047, Abstr., 3839:

> SSAO is a member of the amine oxidase family of enzymes which includes monoamine oxidases A and B and polyamine oxidase in addition to SSAO. *The observation that site selective deuteration can have a marked effect on the metabolic stability of drug molecules metabolized by enzymes of this family is quite well known*.

(emphasis added, internal references omitted).

Patent Owner attempts to discredit these teachings by characterizing the teachings of Kong as "hypothetical," arguing that other enzymes might be involved, and noting that transaminase is not a "natural" substrate of GABA transaminase. We are not persuaded. Kong's suggestions are backed by credible experimental data (*see, e.g.*, Ex. 1005, Example 5), and the catabolism of drugs in the body by enzymes is hardly unknown in the art. We find that a skilled artisan would have understood that Kong and the prior art credibly teach that MAO and GABA transaminase catalyze the degradation of tramiprosate to 2-CES *via* oxidative deamination to the carboxylic acid.

Similarly, with respect to assumptions (b)–(i), we acknowledge that there are no certainties, and that there is a degree of unpredictability, in all drug development. Nevertheless, a certainty of success is not required, only an expectation of success that is reasonable. *Longi*, 795 F.2d at 897. Essentially, Patent Owner points out a list of possible "what ifs?" that are, for the reasons we have explained, outbalanced by the teachings and suggestions of the prior art, which suggest a *reasonable* expectation of success. As Schofield teaches:

> The observation that site selective deuteration can have a marked effect on the metabolic stability of drug molecules metabolized by enzymes of this family is quite well known. For example, the deuterated analogue of the MAO-inhibitor phenelzine **25** had increased biological activity in rats compared to the non-deuterated parent. It was further shown that **25** is catabolized by the same enzyme. Similarly, it has recently been proposed that administration of deuterated lysine **26** could potentially be beneficial in pathologies where lysyl oxidase (LOX), a similar enzyme to SSAO, is implicated. The basis of this hypothesis is that the high observed KIE for oxidative deamination of **26**, a key step involved in LOX biosynthesis *in vivo*, will modulate the amount of LOX present in the organism, reducing its nefarious effects.
>
> Thus, our observation of a marked in vitro effect of deuteration on the rate of oxidative deamination of [deuterated AVE5638] is in accordance with these observations for the amine oxidase family as a whole. *This has been explained by a common mechanism in which H abstraction by the enzyme of an amine's α hydrogen atom is the rate limiting step*.

Ex. 2047, 3839 (internal references omitted, emphasis added).

Indeed, some of Patent Owner's arguments cut against its own thesis. For example, with respect to assumption (g), Patent Owner contends that Table 8 of the Specification of the '323 patent demonstrates that ^{18}O substitution of the phenylalanine prodrug of tramiprosate reduces prodrug cleavage dramatically relative to di-deuteration at the third carbon, with a concomitant reduction in drug exposure. *Id.* This actually supports Petitioner's contention that deuterium substitution at a site proximal to the enzymatic attack on tramiprosate, e.g., the third carbon, is the most effective site for isotopic substitution by maximizing the KIE of tramiprosate at that site, a contention that is supported by Schofield. *See* Ex. 2047, 3839.

Having evaluated the parties' arguments and the evidence of record, we conclude that Petitioner has demonstrated by a preponderance of the evidence that that the '323 patent is obvious over the prior art cited in Grounds 1 and 2.

H. Ground 3: Patentability of claims 3 and 19 over Kong and Czarnik

1. Petitioner's arguments

Petitioner argues, with respect to claim 3 (which includes tramiprosate) that the claim would have been obvious to a person of ordinary skill in the art in view of the teachings of Czarnik and Kong. Pet. 62 (citing Ex. 1002 ¶¶ 184–188). Petitioner asserts that Czarnik teaches "deuterium-enriched tramiprosate" to be used in "a method for treating Alzheimer's disease." *Id.* (quoting Ex. 1007 ¶¶ 4, 6). Petitioner repeats the arguments made with respect to claim 1 above, and contends that a skilled artisan would be most likely to modify Czarnik's compound 6 to form tramiprosate with substitution only at the third carbon (i.e., R is deuterium). Petitioner argues that a person of ordinary skill in the art would be motivated to make this substitution in view of Kong's teaching that metabolism occurs at the third carbon, and includes breaking the carbon-hydrogen bonds. *Id.* (citing Ex. 1002 ¶¶ 71–78).

Petitioner argues that a skilled artisan would also be motivated to select ^{15}N for X, in view of Kong's teaching that the amino group is removed during metabolic degradation of tramiprosate. Pet. 62. According to Petitioner, such an artisan would therefore be motivated to replace a natural abundance nitrogen with ^{15}N in order to increase the strength of the N-C bond and slow the rate of metabolic degradation of tramiprosate. *Id.*

With respect to claim 19, which recites tramiprosate, Petitioner argues that this claim, too, would have been obvious to a person of ordinary skill in the art in view of the teachings of Czarnik and Kong. Pet. 63 (citing Ex. 1002 ¶¶ 189–190). Petitioner contends that a skilled artisan would have been motivated to modify compound 6 of Czarnik to have deuterium at the third carbon in view of Kong's teachings concerning tramiprosate metabolism. *Id.* Petitioner again argues that Kong teaches that the sole metabolite in mammals of tramiprosate is 2-CES. *Id.* (citing Ex. 1005, col. 2, ll. 15–54). Petitioner contends that forming 2-CES requires breaking the C-H bonds at the third carbon. *Id.* Therefore, argues Petitioner, substituting deuterium for natural-abundance hydrogen will increase the KIE. *Id.* (citing Ex. 1002 ¶¶ 71–78 (discussing detailed mechanism)).

Petitioner asserts that both claims 3 and 19 feature tramiprosate without addition of a protecting group, and, therefore, that Patent Owner's argument during prosecution that the claims were patentable because Czarnik does not disclose compounds without protecting groups is not applicable to these claims. Pet. 63. Petitioner therefore contends that a person of ordinary skill in the art would have been motivated to substitute deuterium for hydrogen at the third carbon of tramiprosate, as recited in claims 3 and 19. *Id.*

Finally, Petitioner argues that a person of ordinary skill in the art would have had a reasonable expectation of success in preparing isotopically-substituted tramiprosate as recited in claims 3 and 19 in view of the teachings of Czarnik and Kong. Pet. 63–64. Petitioner argues that Czarnik expressly teaches deuterium-substituted tramiprosate, and Kong identifies the positions to substitute to yield the claimed compounds. *Id.* at 64 (citing Ex. 1002 ¶ 191).

In view of these combined teachings, argues Petitioner, a skilled artisan would have had a reasonable expectation of success in achieving the claimed compound. Pet. 64. Petitioner also contends that a person of ordinary skill in the art would have reasonably expected to achieve a functioning drug, since substitution of a heavier isotope is expected to "have negligible steric consequences or influence on physiochemical properties," because Czarnik's isotopically-substituted compounds would be expected to yield improved pharmacokinetics associated with reduced rate of metabolic degradation of tramiprosate. Pet. 34.

2. Patent Owner's arguments

Patent Owner argues that Petitioner fails to establish that a person of ordinary skill in the art would have had reason to modify Czarnik's compound 6 to arrive at any of the compounds encompassed by claims 3 and 19. PO Resp. 44.

According to Patent Owner, Czarnik's compound 6 has two deuterium atoms at each of the three backbone carbons. PO Resp. 4. By contrast, argues Patent Owner claim 3 allows, and claim 19 requires, deuterium substitution at the third carbon, and not at any other carbon. *Id.*

Patent Owner contends that Petitioner's reliance on Kong's teaching that metabolism occurs at the third carbon offers no sound rationale for removing the four deuterium atoms at the first and second carbons in Czarnik compound 6, which its expert indicated would be expected to perform the same or better than compounds only di-deuterated at the third carbon. PO Resp. 45 (citing Ex. 2042 ¶ 277; Ex. 2041, p. 176, ll. 10–17).

Patent Owner also repeats its argument that a skilled artisan would not ignore the stated benefits of tramiprosate prodrugs espoused by Kong and Tolar and forgo a prodrug as would be required to arrive at claims 3 and 19. PO Resp. 45.

3. Analysis

With respect to *Graham* factors (1) and (3), and as we have explained, Kong teaches both tramiprosate (which falls within the scope of claim 3, and its preferred prodrug, Val-APS (which falls within the scope of claim 1), with the exceptions of the requisite isotopic substitution at the third carbon. *See* Section III.E.1, *supra*. Kong also teaches that administration of the prodrug Val-APS results in greater bioavailability of tramiprosate, as demonstrated by increased C_{max} and AUC when compared to administration of tramiprosate. *Id.* Kong also provides experimental data supporting its suggestion that enzymatic degradation of tramiprosate may occur by the action of GABA transaminase or MAOs, and that, in mammals including humans, the principal metabolite of this degradation is 2-CES, formed by deamination and carboxylation at the number 3 carbon. Ex. 1005, col. 2, ll. 10–43, Example 3, Table 10.

Furthermore, Kong expressly teaches that its claimed compositions:

> [A]lso include isotopically labeled compounds where one or more atoms have an atomic mass different from the atomic mass most abundantly found in nature. Examples of isotopes that may be incorporated into the compounds of the present invention include, but are not limited to $^{2}H(D)$, $^{3}H(T)$, ^{18}O, ^{17}O, ^{11}C, ^{13}C, ^{14}C, ^{15}N, etc.

Ex. 1005, col. 7, ll. 46–51.

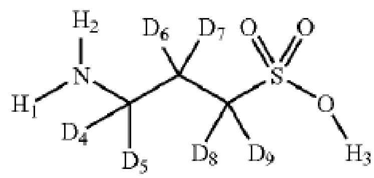
Czarnik teaches isotopic substitution at various loci on the tramiprosate molecule, including at the third carbon. *See* Section III.E.2 *supra*. Specifically, Czarnik encourages substitution of deuterium at the three carbons in the chain of the molecule (including the third carbon) and discourages substitution at the amino and sulfonic acid terminal functional groups, as these hydrogens are easily exchanged. *Id.* Czarnik teaches that substitution can be made at one, several, or all of the hydrogens along the three-carbon backbone of the tramiprosate molecule. *Id.*

In sum, we conclude that the combined teachings of Kong and Czarnik meet all of the limitations of the claims 1 and 3. Specifically, it is uncontested that the references teach the basic chemical structure of the tramiprosate and Val-APS molecules, as recited in the claims. Furthermore, the references expressly teach the requisite isotopic substitutions at the loci recited in the claims. Czarnik expressly teaches substitution of deuterium for natural abundance hydrogen at locations on the carbon backbone of the molecule, including at the CR_2 site recited in the claims. *See* Ex. 1007 ¶ 48. Kong also teaches substitution of isotopes in tramiprosate and Val-APS, including substitution of ^{15}N and ^{13}C and ^{18}O. *See* Kong, col. 4, ll. 46–51. We conclude that a person of skill in the art would have found it obvious to make those substitutions at loci close to the site at which Kong teaches degradative metabolism of tramiprosate or Val-APS occurs, *viz.*, adjacent to the third carbon. *Id.* at col. 2, ll. 39–54

In sum, we conclude that the combined teachings of Kong and Czarnik meet all of the limitations of the claims 1 and 3. Specifically, it is uncontested that the references teach the basic chemical structure of the tramiprosate and Val-APS molecules, as recited in the claims. Furthermore, the references expressly teach the requisite isotopic substitutions at the loci recited in the claims. Czarnik expressly teaches substitution of deuterium for natural abundance hydrogen at locations on the carbon backbone of the molecule, including at the CR_2 site recited in the claims. See Ex. 1007 ¶ 48. Kong also teaches substitution of isotopes in tramiprosate and Val-APS, including substitution of ^{15}N and ^{13}C and ^{18}O. *See* Kong, col. 4, ll. 46–51. We conclude that a person of skill in the art would have found it obvious to make those substitutions at loci close to the site at which Kong teaches degradative metabolism of tramiprosate or Val-APS occurs, *viz.*, adjacent to the third carbon. *Id.* at col. 2, ll. 39–54.

Furthermore, we conclude that a person of ordinary skill in the art would have been motivated to combine the teachings of the references. Substitution of ^{15}N and ^{13}C and ^{18}O isotopes at loci adjacent or proximal to the site of enzymatic degradation (*see* Kong. col. 2, ll. 20–54) would increase the KIE at those sites and minimize the catabolic degradation of the molecule.

We are not persuaded that Patent Owner's arguments overcome Petitioner's showing of unpatentability. As we have explained, we decline to interpret the teachings of Czarnik as narrowly as Patent Owner would have us do. Czarnik's compound 6 is reproduced again below:



Czarnik's compound 6 depicts a molecule of tramiprosate di-deuterated at each carbon of the 3-carbon backbone of the molecule

However, Czarnik expressly teaches, in relation to, *inter alia*, compound 6 that: "numerous modifications and variations of the present invention are possible in light of the above teachings. It is therefore to be understood that within the scope of the appended claims, the invention may be practiced otherwise that as specifically described herein." Ex. 1007 ¶ 48. Furthermore, Czarnik contemplates molecules of tramiprosate with deuterium replacing only a single hydrogen or several:

> If not specifically noted, the percentage of enrichment refers to the percentage of deuterium present in the compound, mixture of compounds, or composition. Examples of the amount of enrichment include from about 0.5, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 16, 21, 25, 29, 33, 37, 42, 46, 50, 54, 58, 63, 67, 71, 75, 79, 84, 88, 92, 96, to about 100 mol %. Since there are 9 hydrogens in tramiprosate, replacement of a single hydrogen atom with deuterium would result in a molecule with about 11% deuterium enrichment. In order to achieve enrichment less than about 11 %, but above the natural abundance, only partial deuteration of one

> site is required. Thus, less than about 11 % enrichment would still refer to deuterium-enriched tramiprosate.

Id. ¶ 15. Extending this logic, Czarnik teaches that:

> In another embodiment, the present invention provides a novel, deuterium enriched compound of formula I or a pharmaceutically acceptable salt thereof, wherein the abundance of deuterium in R_4-R_9 [i.e., at the three-carbon backbone] is at least 17%. The abundance can also be (a) at least 33%, (b) at least 50%, (c) at least 67%, (d) at least 83%, and (e) 100%.

Id. ¶ 24. Di-deuteration at the third carbon would therefore result in abundance of "at least 33%." *See also id.* ¶ 32, claims 4, 10, 16.

Furthermore, we have explained in detail in Section II.G.1 *supra* why a person of ordinary skill in the art, comprehending the teachings of Kong and the related prior art, including Schofield, would be motivated to target the substitution of deuterium for hydrogen at the third carbon, because that is the site of enzymatic attack in the oxidative deamination to carboxylic acid. *See* Ex. 1005, col. 2, ll. 8–54, Ex. 2047, 3839.

Accordingly, we conclude that Petitioner has demonstrated by a preponderance of the evidence that challenged claims 3 and 19 of the '323 patent are obvious over the combined cited prior art that form the basis of Ground 3.

I. Reasonable Expectations of Chemical or Functional Success

Patent Owner next revisits the question of whether a person of skill in the art would have had a reasonable expectation of attaining either the requisite functional or chemical success. PO Resp. 47.

1. Reasonable expectation of functional success

Briefly, Patent Owner contends that, given the unpredictability of the effects of isotopic modifications on pharmacological properties of pharmaceutical compounds, Petitioner has failed to demonstrate a reasonable expectation of functional success. PO Resp. 48 (citing Otsuka, 678 F.3d at 1298). Patent Owner notes that, in another proceeding, the Board found that the unpredictability of deuteration on the metabolic and pharmacologic properties of a drug are unpredictable and did not support Petitioner's theory of a reasonable expectation of success. *Id.* (citing *Neptune Generics, LLC., v. Auspex Pharms., Inc.*, IPR2015-01313, Paper 25, 18 (PTAB Dec. 9, 2015) (denying institution).

Patent Owner argues that the literature demonstrates that a KIE may slow down, speed up, or have negligible effects on metabolism of a particular compound, and that that the direction and magnitude of any KIE can vary significantly based on the identity of the enzyme-substrate Pair. PO Resp. 48–49. As a result, Patent Owner argues, a skilled artisan cannot reasonably predict how a KIE will impact the pharmacological properties of a given compound. *Id.* at 49 (citing Ex. 2041, p. 212, ll. 11–14, p. 214, ll. 5–8; also citing Ex. 2041 ¶¶ 93, 170–172, 173–182).

We acknowledge Patent Owner's argument that predicting the KIE for a given chemical substitution can be an uncertain matter. Nevertheless, the evidence of record suggests that deuterium substitution at a molecular site adjacent to enzymatic attack by enzymes of the monoamine oxidase family strongly suggest that the KIE in that instance assists in resistance to enzymatic degradation. *See* Ex. 2047, 3839 as quoted *supra* in Section III.G.3.b. We find that the studies recited in this passage would have motivated a person of skill in the art to substitute deuterium (or 15N) at the active site of enzymatic attack to exploit the KIE. Again, we remind Patent Owner that there need only be a reasonable expectation of functional success for the skilled artisan. We conclude that the successful studies described in Schofield, combined with the teachings of Kong, would provide such a reasonable expectation.

2. Reasonable expectation of chemical success

Patent Owner next argues that a person of ordinary skill in the art could not have had a reasonable expectation of success in synthesizing tramiprosate or Val-APS di-deuterated at the third carbon. PO Resp. 52. According to Patent Owner, isotopically-modified compounds often require distinct and dedicated syntheses—including specialized reagents, synthetic steps, and/or order of synthetic steps—compared to their non-isotopically-modified analogs. *Id.* (citing Ex. 2041, pp. 199–200, ll. 4–17; Ex. 2042 ¶¶ 66–69; Ex. 2047, 3833–3834. Patent Owner contends that these dedicated syntheses often require complex design to ensure introduction of a desired isotope at an intended position and to the exclusion of other positions. *Id.* (citing Ex. 2047 ¶¶ 66–69).

Patent Owner asserts that obtaining di-deuterated forms of tramiprosate would be particularly problematic, because procedures to introduce isotopic substitutions could result in two deuterium atoms on the desired carbon (e.g., C3), on a different carbon (e.g., C1, C2), or two different carbons, or could result in over- or under-deuteration (e.g., tri-deuteration or mono-deuteration). PO Resp. 52 (citing Ex. 2043 ¶ 129).

Patent Owner argues that neither Kong, Czarnik, nor Tolar teaches the skilled artisan how to address these many challenges, and maintains that the knowledge of a person of skill in the art would be insufficient to supplement these gaps. PO Resp. 53 (citing Ex. 2042 ¶¶ 225–227). Patent Owner notes that Petitioner's declarant acknowledged that a person of ordinary skill in the art might not have experience preparing isotopically-modified compounds. *Id.* (citing Ex. 2041, p. 27, ll. 11–14).

We do not find Patent Owner's arguments persuasive in view of the teachings of the prior art. For example, Naicker[13] teaches that:

> Deuterated compounds have been used in pharmaceutical research to investigate the *in vivo* metabolic fate of the compounds by evaluation of the mechanism of action and metabolic pathway of the non deuterated parent compound.
> ….
> Incorporation of a heavy atom particularly substitution of deuterium for hydrogen, can give rise to an isotope effect that could alter the pharmacokinetics of the drug.

Ex. 1017 ¶¶ 53–54. Furthermore, Schofield discusses a number of studies in which deuterium is substituted for hydrogen at the site adjacent to MAO family enzymatic attack to exploit the KIE, and expressly suggests that such substitution might be a useful method for blocking amine oxidase mediated oxidative deamination. Ex. 2047, 3839. Petitioner's Declarant, Dr. Guengerich specifically points out that "Schofield does not support Dr. Sessler's opinion that deuterated compounds are particularly difficult to

[13] Naicker et al., (US 2012/0214745 A1, August 23, 2012) ("Naicker") Ex. 1017.

synthesize." Ex. 1046 ¶ 77. Dr. Guengerich continues, referring to Patent Owner's contention that he acknowledged the difficulty of synthesizing D_2-tramprosate:

> I do not believe that it would be difficult for a POSA to obtain D_2-tramiprosate, nor have I seen evidence in the [Patent Owner's Response] or Dr. Sessler's Declaration indicating to me that it would be difficult. D_2-tramiprosate may be obtained by synthesizing from isotopically-substituted precursors, as stated in Czarnik. These precursors are "specialized" in the sense that they are isotopically-substituted, rather than natural abundance compounds. However, they are not necessarily rare or unusual. Indeed, many isotopically-substituted precursors are commercially available. This is the sense in which I used "specialized."

Id. ¶ 79.

We conclude that the preponderance of the evidence does not support Patent Owner's contention that a person of ordinary skill in the art would not have had a reasonable expectation of chemical success in synthesizing D_2-tramiprosate or the D_2-Val-APS prodrug.

J. Secondary Indicia of Nonobviousness

1. Unexpected Results

Finally, Patent Owner argues that deuterated tramiprosate prodrugs, including D_2-Val-APS, exhibit unexpectedly beneficial pharmacokinetic properties. PO Resp. 56. Patent Owner points to data disclosed by the '323 patent as demonstrating that D_2-Val-APS exhibits superior pharmacokinetic ("PK") properties. Patent Owner contends that Figure 1 of the '323 patent shows that D_2-Val-APS "improved plasma drug exposure significantly" in mice, demonstrating a more than 2-fold increase of C_{max} for the drug concentration relative to D_2-tramiprosate. PO Resp. 56–57 (quoting Ex. 2042 ¶ 281; and citing Ex. 1001, col. 45, ll. 36–41, FIG. 1). Patent Owner also points to Table 7, which Patent Owner contends shows that D_2-Val-APS exhibited a 24% increase in AUC and a 40% increase in C_{max} relative to D_2-tramiprosate in rats, with even larger improvements relative to tramiprosate. *Id.* at 57 (citing Ex. 1001, col. 46, ll. 25–35, Table 7; Ex. 2042 ¶ 281. Patent Owner argues further that D_2-Val-APS was also "converted easily" to D_2-tramiprosate in the subject after administration. *Id.* (citing Ex. 1001, col. 45, ll. 36–41).

"[W]hen unexpected results are used as evidence of nonobviousness, the results must be shown to be unexpected compared with the closest prior art." *In re Baxter Travenol Labs.*, 952 F.2d 388, 392 (Fed. Cir. 1991). The closest prior art to the '323 patent is Kong, which teaches both tramiprosate and its prodrug Val-APS (we again emphasize that the difference between tramiprosate and D_2-tramiprosate, and between Val-APS and D_2-Val-APS, is a total of 2 neutrons). Specifically, in Example 3, Kong teaches that a "2-fold increase in plasma concentration (C_{max}) of tramiprosate was observed when orally administering [Val-APS] compared to [tramiprosate]." Ex. 1005, col. 110, ll. 12–14; *see also* Table 10. In other words, in Kong, administration of the prodrug Val-APS showed a similar increase in plasma C_{max} of tramiprosate compared to administration of tramiprosate, as D_2-Val-APS showed in comparison to D_2-tramiprosate. Given the teachings of Kong as the closest prior art, a person of ordinary skill in the art would not be surprised that the deuterated Val-APS increased the concentration of circulating deuterated tramiprosate to the same degree as non-deuterated

Val-APS did with respect to non-deuterated tramiprosate. The effect is the same.

What Patent Owner does *not* show is any comparison between the effect of administering D_2-Val-APS *versus* Val-APS. Only by such a comparison can the effect of *deuteration* of the prodrug be established. Absent any such direct comparison between the claimed compositions and the closest prior art, Patent Owner's attempt to demonstrate unexpected or surprising properties of the claimed D_2-Val-APS founders.

Patent Owner also argues that the '323 Patent discloses that D_2-tramiprosate exhibited surprisingly superior pharmacokinetic properties compared to tramiprosate. PO Resp. 61 (citing Sessler ¶¶ 280–81). Patent Owner points to Figure 1 of the '323 patent, which Patent Owner argues demonstrates that D_2-tramiprosate "improved plasma drug exposure significantly" in mice compared to administration of tramiprosate. *Id.* (citing Ex. 1001, col. 45, ll. 36–41, Fig. 1). Figure 1 of the '323 patent is reproduced below:

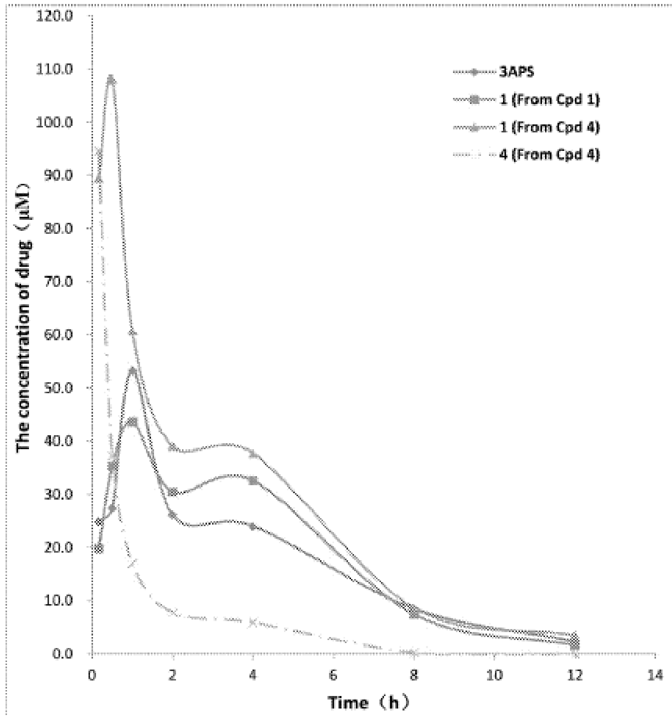


Figure 1 depicts plasma concentration-time curves of the compound following an oral administration of 3APS) (nonisotope-enriched tramiprosate), compound 1 (D_2-tramiprosate), and compound 4 (D_2-Val-APS), including the concentration-time curve of compound 4 in the same experiment.

Patent Owner also points to Table 7 as showing that D_2-tramiprosate exhibited a 14% increase in AUC and a 23% increase in C_{max} in rats relative to tramiprosate. PO Resp. 61 (citing Ex. 1001, Table 7; Ex. 2042 ¶ 281). Patent Owner asserts that, because the impacts of isotopic substitution on pharmacological properties are unpredictable, these data provide unexpected results. *Id.* (citing Sessler ¶¶ 280–81).

We have already concluded that the increased C_{max} in D_2-Val-APS (compound 4[14]) compared to D_2-tramiprosate (compound 1[15]) or tramiprosate would not be surprising to a skilled artisan in view of the teachings of Kong, the closest prior art. *See* Section II.G.1. The relevant remaining question, then, is whether D_2-tramiprosate exhibits surprising results compared to administration of tramiprosate. The '323 patent discloses, with respect to Figure 1, that "[t]he results indicate that at the mole equivalent oral dose, the isotope-enriched compounds 1 and 4 improved plasma drug exposure significantly." Ex. 1001, col. 45, ll. 36–38. Table 7 of the '323 patent is reproduced below:

TABLE 7

PK parameters for 3APS (of natural abundance), compound 1, and compound 4 in SD rats.

Parameter	Unit	3APS	Compound 1 (following administration of) 1	4
AUC (0-t)	µg · h/L	47212	53916	66900
$t_{1/2}$	h	3.7	3.7	2.4
T_{max}	h	0.5	0.5	0.4
C_{max}	µg/L	18707	23017	32216

<u>Table 7 depicts the pharmacokinetic results from Figure 1</u>

Figure 1 and Table 7 demonstrate that administration of D_2-tramiprosate (compound 1) results in an increased C_{max} and AUC when compared to administration of non-deuterated tramiprosate (APS). The

[14] *See* Ex. 1001, Table 1.

[15] *Id.*

question before us is whether these results would have been surprising to a person of ordinary skill in the art.

We conclude that they would not. As an initial matter, unexpected results that are probative of nonobviousness are those that are "different in kind and not merely in degree from the results of the prior art." *Iron Grip Barbell Co. v. USA Sports, Inc.*, 392 F.3d 1317, 1322 (Fed. Cir. 2004) (citation omitted). It is undeniable that the deuterated tramiprosate shows an increase in C_{max} and AUC compared to non-deuterated tramiprosate, but the increases in C_{max} (19%) and AUC (14%) are matters of degree, and not of kind. This is particularly true of the difference in AUC, which documents the total amount of plasma drug over time.

More importantly, we do not find these results unexpected in view of the teachings of the prior art, particularly Czarnik and Schofield, and the general knowledge in the field of the KIE. *See, e.g.*, Ex. 2047, 3831:

> The theoretical principle underlying these attempts, that of the kinetic isotope effect (KIE), relies on the assumption that a metabolic reaction which breaks a C–H bond in the rate-determining step will be slowed down when hydrogen is substituted for deuterium. Depending on the mode of action, this could translate to increased exposure to the parent, decreased formation of a toxic or reactive metabolite, or switching to a metabolic pathway which enhances formation of a species with increased activity.

(internal references omitted). We acknowledge that there is a degree of unpredictability in the KIE, nevertheless, and as we have explained *supra*, the teachings of the prior art are such that those of skill in the art would have reasonably expected that deuteration of the third carbon of tramiprosate would significantly increase the C_{max} and AUC of D_2-tramiprosate compared to tramiprosate. Patent Owner's argument that its claimed composition exhibits unexpected results are consequently not persuasive.

2. Alleged Failure of Others

Finally, Patent Owner argues that the failure of others in the art to improve pharmacological properties of drugs by isotopic substitution provides further objective evidence of the nonobviousness of the challenged claims. PO Resp. 67 (citing Ex. 2042 ¶¶ 201, 294–98). According to Patent Owner, prior to the '323 Patent, no deuterated drugs had been approved, and to this day, only one such compound has received FDA approval. *Id.* Patent Owner contends that the prior art included numerous reports of the failure of isotopically-modified pharmaceuticals. *Id.* (citing Ex. 2042 ¶ 295; Ex. 2041, p. 232, ll. 13–21).

We do not find this argument persuasive. Patent Owner adduces no evidence that shows specifically that attempts to synthesize or use D_2-tramiprosate or the D_2-Val-APS prodrug resulted in failure, despite the express teachings and suggestions of Czarnik and Kong. Moreover, and as we have explained, Schofield provides an express demonstration of successes in inhibiting the same enzymatic deamination to carboxylic acid by amine oxidase reaction that is believed to be the mechanism by which tramiprosate is degraded to 2-CES. *See* Ex. 2047, Abstr., 3839.

We acknowledge that there were, at the time of filing, no deuterated drugs approved by the FDA for pharmaceutical use. But FDA approval is not the sole standard by which success or failure is measured. The prior art forming the bases of Grounds 1–3, as well as other cited prior art, indicate that investigations concerned with employing the KIE in developing drugs was very much ongoing at the time of filing. *See, e.g.*, Ex. 2047, 3839; Ex. 1013[16], 403; Ex. 1017 ¶¶ 44–45. Indeed, Dr. Guengerich testified that, as of March 2017, Teva[17] announced that deuterabenazine had demonstrated success in clinical trials, and other successful examples were known in the art. Ex. 1046 ¶ 116 (citing, e.g., Ex. 1018[18]; Ex. 1043[19]; Ex. 1012[20]).

In summary, we conclude that Patent Owner's arguments with respect to secondary indicia of nonobviousness are insufficient to overcome Petitioner's showing, by a preponderance of the evidence, that claims 1–20 of the '323 patent are obvious over the prior art cited in Grounds 1–3.

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[16] L.E. Dyck et al., *Effects of Deuterium Substitution on the Catabolism of P-Phenylethylamine: An In Vivo Study*, 46(2) NEUROCHEM. 400–04 (1986) ("Dyck") Ex. 1013.

[17] Teva Pharm. Indus. Ltd., *Teva Announces Positive Top-Line Data from Second Phase III Study of SD-809 in Tardive Dyskinesia (TD)*, September 22, 2016 ("Teva") Ex. 1065.

[18] STABLE ISOTOPES IN PHARMACEUTICAL RESEARCH, PHARMACOCHEM LIB. VOL. 26 (T.R. Browne, ed.) (Elsevier 1997) ("Browne") Ex. 1018.

[19] S.L. Harbeson et al., *Deuterium Medicinal Chemistry: A New Approach to Drug Discovery and Development*, 2 MEDCHEM NEWS 98–22 (May 2014) ("Harbeson") Ex. 1043.

[20] A.J. Morgan et al., *Old Drugs Yield New Discoveries: Examples from the Prodrug, Chiral Switch, and Site-Selective Deuteration Strategies*, in DRUG REPOSITIONING: BRINGING NEW LIFE TO SHELVED ASSETS AND EXISTING DRUGS, 1st Ed. (M. J. Barratt et al., eds.) 291–343 (J. Wiley & Sons, 2012) ("Morgan") Ex. 1012.

K. Conclusion

For the foregoing reasons, and after having analyzed the entirety of the record and assigning appropriate weight to the cited supporting evidence, we determine that Petitioner has established by a preponderance of the evidence that claims 1–17 and 19–20 are unpatentable under 35 U.S.C. § 103 as being obvious over Kong and Czarnik (Ground 1), claims 1–20 and Tolar are unpatentable under 35 U.S.C. § 103(a) as being obvious over Kong, Czarnik, and Tolar (Ground 2), and claims 3 and 19 are unpatentable under 35 U.S.C. § 103 as being obvious over Czarnik and Kong (Ground 3).

IV. PATENT OWNER'S CONTINGENT MOTION TO AMEND

A. Introduction

Contingent on the determination that claim 13 is unpatentable, Patent Owner requests that we replace claims 13–14 with proposed corresponding substitute claims 21–22. *See* Mot. Amend 1–2. Additionally, Patent Owner requests that, should the Board find claim 19 unpatentable, we replace claim 19 with claim 23. *Id.* As discussed in Section III, we determine that Petitioner has demonstrated by a preponderance of the evidence that claims 13 and 19 are unpatentable. We therefore consider Patent Owner's Contingent Motion to Amend ("MTA").

B. Pilot Program Participation

A pilot program for motion to amend practice and procedures became available to all proceedings instituted on or after March 15, 2019. *See Notice Regarding a New Pilot Program Concerning Motion to Amend Practice and Procedures in Trial Proceedings Under the America Invents*

Act Before the Patent Trial and Appeal Board, 81 FED. REG. 9497 (March 15, 2019) (the "Pilot Program"). Pursuant to the Pilot Program, a patent owner may request, in its motion to amend, that the Board issue preliminary guidance after the petitioner files its opposition to the motion to amend. *See id.* at 9499, 9500. Preliminary guidance on a motion to amend is not binding on the Board. *See id.* at 9500. After receiving preliminary guidance from the Board, a patent owner may elect to file a revised motion to amend, file a reply to petitioner's opposition and/or the preliminary guidance, or take no action. *See id.*

C. Principles of Law

In an *inter partes* review, amended claims are not added to a patent as a matter of right, but rather must be proposed as a part of a motion to amend. 35 U.S.C. § 316(d). "Before considering the patentability of any substitute claims, . . . the Board first must determine whether the motion to amend meets the statutory and regulatory requirements set forth in 35 U.S.C. § 316(d) and 37 C.F.R. § 42.121." *Lectrosonics, Inc. v. Zaxcom, Inc.*, IPR2018-01129, Paper 15 at 4 (PTAB Feb. 25, 2019) (precedential). Accordingly, we must consider whether: (1) the amendment proposes a reasonable number of substitute claims; (2) the proposed claims are supported in the original disclosure (and any earlier filed disclosure for which the benefit of filing date is sought); (3) the amendment responds to a ground of unpatentability involved in the trial; and (4) the amendment does not seek to enlarge the scope of the claims of the patent or introduce new subject matter. *See* 35 U.S.C. § 316(d); 37 C.F.R. § 42.121.

The Board must assess the patentability of proposed substitute claims "without placing the burden of persuasion on the patent owner." *Aqua Prods., Inc. v. Matal*, 872 F.3d 1290, 1328 (Fed. Cir. 2017) (en banc); *see also Lectrosonics, Inc. v. Zaxcom, Inc.*, IPR2018-01129, Paper 15 at 3–4 (PTAB Feb. 25, 2019) (precedential). Subsequent to the issuance of *Aqua Products*, the Federal Circuit issued a decision in *Bosch Automotive Service Solutions, LLC v. Matal*, 878 F.3d 1027 (Fed. Cir. 2017) ("*Bosch*"), as well as a follow-up Order amending that decision on rehearing. *See Bosch Auto. Serv. Sols., LLC v. Iancu*, Order on Petition for Panel Rehearing, No. 2015-1928 (Fed. Cir. 2018).

In accordance with *Aqua Products*, *Bosch*, and *Lectrosonics*, Patent Owner does not bear the burden of persuasion to demonstrate the patentability of the substitute claims presented in the motion to amend. Rather, ordinarily, "the petitioner bears the burden of proving that the proposed amended claims are unpatentable by a preponderance of the evidence." *Bosch*, 878 F.3d at 1040 (as amended on rehearing); *see Lectrosonics*, Paper 15 at 3–4. In determining whether a petitioner has proven unpatentability of the substitute claims, the Board focuses on "arguments and theories raised by the petitioner in its petition or opposition to the motion to amend." *Nike, Inc. v. Adidas AG*, 955 F.3d 45, 51 (Fed. Cir. 2020).

D. Analysis

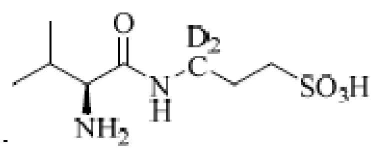
We consider Patent Owner's Contingent Motion to Amend and Petitioner's Opposition, along with the subsequently filed Reply and

Sur-reply. We begin our analysis by addressing the statutory and regulatory requirements for a motion to amend, followed by addressing the Petitioner's assertions of unpatentability of proposed substitute claims 21–23.

1. Patent Owner's Substitute Claims 21–23

Patent Owner proposes changes to claims 13, 14, and 19 in substitute independent claims 21, and 22 (corresponding to claims 13 and 14) and 23 (corresponding to claim 19). Mot. Amend. 1, A1–A3. Proposed substitute claims 2, 22, and 23 are reproduced below. New limitations are italicized.

21. The compound of claim 1, which is



Mot. Amend. A–1. Proposed substitute claim 22 recites:

> 22. The compound of claim 21, wherein the level of isotope enrichment in the compound is about 95% or more.

Id. at A–3. Proposed substitute claim 23 recites:

> 23. A compound which is
>
> 
>
> *wherein the level of isotope enrichment in the compound is about 95% or more.*

Id.

2. Reasonable Number of Proposed Substitute Claims

We agree with Patent Owner that the proposed substitution of a single claim for each of challenged claims 13, 14, and 19 meets the requirement for a reasonable number of substitute claims. *See* Mot. Amend 4; 37 C.F.R. § 42.121(a)(3) (establishing a rebuttable presumption that only one substitute claim is needed to replace each challenged claim).

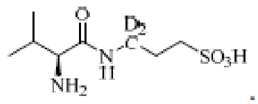
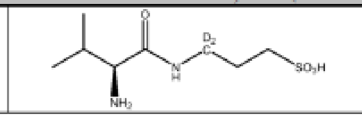
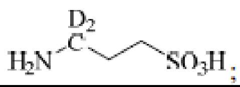
3. Enlargement of Claim Scope

Patent Owner asserts that proposed substitute claims 21–23 do not seek to enlarge the scope of the originally issued claims because each substitute claim narrows the elements of the corresponding original claim and, in claim 23, the added limitation does not broaden the scope. *See* Mot. Amend 3–4. Petitioner does not dispute Patent Owner's contention that the proposed substitute claims do not seek to enlarge the scope of the claims of the '323 Patent. *See generally* Opp.

Based on the record before us, we agree with Patent Owner, and determine that the proposed substitute claims meet the requirements of 37 C.F.R. § 42.121(a)(2). The proposed substitute claims delete certain broadening limitations, and substitute claim 23 adds a single limitation that narrows the scope of the substitute claims.

4. New Matter/35 U.S.C. § 112 Written Description Support

Patent Owner asserts that the narrowing limitations of the proposed substitute claims are supported and enabled by the Specification of the '255 Application, parent to the '323 Patent, and particularly point out and distinctly claim the invention. *See* Rev. Mot. Amend 5–8. Patent Owner provides the following chart, indicating where each limitation receives support in the Specification:

Claim Limitation	Support in U.S.S.N. 15/476,255 (EX2066)
21. The compound of claim 1, which is  .	4  EX2066, [0023], Table 1; *see also* claim 27, FIGs. 1-2.
22. The compound of claim 21,	*See supra* for claim 21.
wherein the level of isotope enrichment in the compound is about 95% or more.	"In still another embodiment, the level of isotope-enrichment in an isotope-enriched compound of the invention (e.g., 3APS, a compound of any of Formulae (1)-(VI), etc.) is about 95% or higher, or 100%." EX2066, [0094]; *see also* claim 30.
23. A compound which is  ;	1  EX2066, [0023]; *see also* claim 27, FIGs. 1-2.
wherein the level of isotope enrichment in the compound is about 95% or more.	"In still another embodiment, the level of isotope-enrichment in an isotope-enriched compound of the invention (e.g., 3APS, a compound of any of Formulae (1)-(VI), etc.) is about 95% or higher, or 100%." EX2066, [0094]; *see also* claim 30.

Petitioner does not dispute Patent Owner's statements, and we conclude that the proposed substitute claims do not add new matter. We also conclude that the Specification of the '323 patent provides sufficient written description support and enablement for the limitations of proposed substitute claims 21–23.

5. Responding to a Ground of Unpatentability

Patent Owner asserts that each of substitute claims 21–23 is responsive to a ground of unpatentability involved in this proceeding. Mot. Amend. 3 (citing 37 C.F.R. § 42.121(a)(2)(i)). Specifically, Patent Owner argues that Petitioner in this proceeding has asserted that the asserted prior art renders obvious original claims 13 and 14 (as well as original claim 1, from which claims 13 and 14 ultimately depend) and claim 19, which this Motion to Amend contingently seeks to amend. *Id.*

Patent Owner argues further that the substitute claims correspond to the original claims and are amended only to remove broadening features, add narrowing features, and alter dependency of the claims. *Id.*

We agree with Patent Owner that the substitute claims are responsive to at least one of the Grounds of unpatentability. *See* Prelim. Guid. 4.

6. Patentability of the Proposed Substitute Claims

Petitioner contests the patentability of the proposed substitute claims based on the same Grounds 1–3 as presented in the Petition. *See* Opp. 6–16.

a. Grounds 1 and 2: Obviousness of Proposed Substitute Claims 21–23 over Kong and Czarnik (Ground 1) and Kong, Czarnik, and Tolar (Ground 2)

We first address Petitioner's arguments regarding the proposed substitute claims. Then we turn to Patent Owner's Reply and Petitioner's Sur-Reply to the Motion to Amend before analyzing the parties' arguments and evidence.

i. Proposed substitute claim 21

Petitioner notes that claim 21 is directed to D_2-Val-APS. Opp. 6. Petitioner essentially repeats its arguments from the Petition, which we already set forth in Sections III.E.4–5 of our Decision *supra*. *See* Opp. 6–9, 12–13.

ii. Proposed substitute claim 22

Petitioner argues that it would have been obvious to a person of ordinary skill in the art to select a level of isotope enrichment in the compound that is about 95% or more. Opp. 9. Petitioner points to the Reply Declaration of Dr. Guengerich, who explains, that it has long been known that a high degree of substitution is desirable. *Id.* (citing Ex. 1046 ¶ 119). Petitioner asserts that numerous studies demonstrate isotopic substitution at or above 95%. *Id.* at 10, 13–14 (citing, e.g., Ex. 1046 ¶ 119; Ex. 1071[21], 1155, n.13; Ex. 1033[22], 13673; Ex. 1009[23]; Ex. 1006[24] ¶ 29).

Petitioner also argues that the ’323 Patent does not actually disclose what degree of deuteration was achieved for D_2-Val-APS or D_2-tramiprosate, nor does it indicate that a certain degree is necessary. Opp. 10 (citing Ex. 1046 ¶ 120; Ex. 1001, *generally*). Petitioner asserts that the ’323 patent discloses enrichment of only a final product for ^{18}O enrichment—and that enrichment is less than 95%. *Id.* (citing Ex. 1001, col. 39, ll. 39–40; col. 40, ll. 19–20; col. 41, l. 1; col. 41, ll. 60–61). To the contrary, Petitioner

[21] K.A. Kurtz et al., *pH and Secondary Kinetic Isotope Effects on the Reaction of D-Amino Acid Oxidase with Nitroalkane Anions: Evidence for Direct Attack on the Flavin by Carbanions*, 119 J. AM. CHEM. SOC. 119, 1155–56 (1997) (“Kurtz”) Ex. 1071.

[22] J.R. Miller et al., *Structure-Activity Relationships in the Oxidation of Para-Substituted Benzylamine Analogues by Recombinant Human Liver Monoamine Oxidase A*, 38 BIOCHEM. 13670–83 (1999) (“Miller”) Ex. 1033.

[23] Ito et al. (US 7,517,990 B2, April 14, 2009) (“Ito”) Ex. 1009.

[24] Gant (US 2010/0120744 Al, May 13, 2010) (“Gant”) Ex. 1006.

argues, the ’323 patent states that “an ‘isotope-enriched’ compound or derivative possesses a level of an isotopic form that is higher than the natural abundance of that form” and may be as low as 2%. *Id.* (citing Ex. 1001, col. 23–24, ll. 35–5).

iii. Proposed substitute claim 23

Petitioner notes that proposed substitute claim 23 is directed to D_2-tramiprosate. Opp. 11. Petitioner essentially repeats the arguments in its Petition, which we already set forth in Sections III.E.4–5 of our Decision *supra*. *See* Opp. 11–12, 14.

b. Ground 3: Obviousness of proposed substitute claim 23 over Czarnik and Kong

i. Proposed substitute claim 23

Only proposed substitute claim 23 is relevant to Ground 3. Petitioner asserts that proposed substitute claim 23 is directed to D2-tramiprosate “wherein the level of isotope enrichment in the compound is about 95% or more.” Opp. 15. Petitioner argues that D_2-tramiprosate would have been obvious in view of Czarnik and Kong, and also obvious to select a level of isotope enrichment that is about 95% or more, for the same reasons that we have already set forth in Section III.H.1. and IV.6.a.ii *supra*. *Id.* at 15–16.

7. Patent Owner’s Reply

In its Reply to Petitioner’s Opposition to the MTA, Patent Owner argues that Petitioner: (1) fails to establish that proposed substitute claims 21–23 are obvious over the cited prior art in Grounds 1–3; and (2) that

Petitioner has failed to rebut Patent Owner's demonstration of unexpected results with respect to the claimed compounds. Opp. Reply 2, 8.

a. Alleged nonobviousness of claims 21–23

Patent Owner argues that Petitioner failed to meet its burden to show that a person of ordinary skill in the art would have been motivated to modify tramiprosate, D_2-Val-APS, or Czarnick's Compound 6, to make D_2-Val-APS or D_2-tramiprosate and would have had a reasonable expectation of being able make these compounds with an isotopic enrichment of about 95% or more. Opp. Reply 3.

Patent Owner contend that Petitioner's reliance upon the expert testimony of its Declarant, Dr. Guengerich, amounts to more than conclusory attorney argument that a skilled artisan would have been motivated, in view of the teachings of the prior art, to synthesize D_2-Val-APS or D_2-tramiprosate or to have a reasonable expectation of success in so doing. Opp. Reply 3. Patent Owner again points to the nine allegedly erroneous assumptions upon which Petitioner's arguments rely to establish motivation, arguing that even a single erroneous assumption would be fatal to Petitioner's case.[25] *Id.* at 4.

By way of example, Patent Owner again argues that a skilled artisan would have avoided di-deuterating the third carbon of Val-APS to avoid

[25] We note that Patent Owner argues that these assumptions are described in the MTA, whereas they are actually argued in the Patent Owner's Response. *See* PO Resp. 27–41. We take this to be a harmless attributional error on the part of the Patent Owner.

causing undesirable changes to its metabolism. Opp. Reply 4. Patent Owner points to Dr. Sessler's expert speculation that such changes could result in the intact prodrug being cleared from the body, so that the active never reaches the brain, or residence times in different physiological locations (e.g., gut, liver) relative to unmodified tramiprosate. *Id.* (citing Ex. 2042 ¶¶ 243–44; Ex. 2041, pp. 171–172, ll. 3–8, p. 172, ll. 15–21).

Patent Owner also contends, again, that Petitioner has failed to establish a reasonable expectation of chemical success in synthesizing either of the compounds recited in proposed substitute claims 12–23. Opp. Reply 6. Notably, Patent Owner argues that Petitioner's Opposition contains no specifics regarding potential synthetic routes available to a person of skill in the art for synthesizing D_2-Val-APS or D_2-tramiprosate. *Id.* at 7. According to Patent Owner, to the degree that Dr. Guengerich proposed any specific synthetic steps within the block-cited testimony, his Declaration provides only a hindsight reconstruction using a synthetic route provided in the '323 patent. *Id.* (citing Opp. 17 (citing Ex. 1046 ¶¶ 72–87); Ex. 1046 ¶¶ 81–86; cf. Ex. 1001, cols. 34–35, ll. 59–36 (Example 1)). Patent Owner argues that, absent hindsight, a person of skill in the art could not have reasonably expected to be able to successfully di-deuterate at the third carbon. *Id.* at 8.

b. Unexpected results

Patent Owner again argues that Petitioner's contention that the compositions recited in claims 21–23 are obvious over the cited prior art is overcome by Patent Owner's demonstration of unexpected results. Opp. Reply. 8. Patent Owner argues that: (1) Petitioner failed to rebut Patent Owner's evidence that Patent Owner provided the proper comparison to what one of skill in the art would have considered to be the closest prior art; and (2) Patent Owner's evidence demonstrates superior performance to what would have reasonably expected by a person of skill in the art. *Id.* at 9. (citing Ex. 2042 ¶¶ 288–292; *In re Merch.*, 575 F.2d 865, 869 n.8 (C.C.P.A. 1978)).

Finally, Patent Owner argues that Czarnik's compound 6 cannot be the closest prior art, as Petitioner allegedly asserts. Opp. Reply 11 (citing Opp. 8). Furthermore, argues Patent Owner, Petitioner adduces no evidence that that Compound 6 had been made by March 2017, as admitted by Dr. Guengerich. *Id.* (citing Ex. 2111, pp. 215–216, ll. 19–11, pp. 217–218, 23–2.

8. Analysis

We are not persuaded that Patent Owner's arguments overcome Petitioner's demonstration, by a preponderance of the evidence, that proposed substitute claims 21–23 are unpatentable under 35 U.S.C. § 103 as being obvious over the prior art cited in Grounds 1–3. Indeed, the fact that the proposed substitute claims recite the same compounds, D_2-tramiprosate and its D_2-Val-APS prodrug, that were principally argued in Petitioner's Brief and Patent Owner's Response, compels the same conclusions, *viz.*, that the proposed claims are unpatentable as being obvious over the cited prior art and that the secondary evidence of nonobviousness is not sufficient to overcome the conclusion of obviousness.

With respect to Petitioner's demonstration that the claims are obvious over the prior art, we have explained with respect to the original claims why we are not persuaded by Patent Owner's arguments with respect to a skilled artisan's motivation and reasonable expectation of success, functional or chemical. With respect to the latter, we find Dr. Sessler's opinions as to the difficulties involved in synthesis to be largely speculative in nature, and at odds, as we have explained, at odds with the teachings of the prior art. *See, e.g.*, Section III.I.2, *supra*. Furthermore, we particularly find Patent Owner's argument that deuterating the prodrug Val-APS might somehow interfere with conversion to D_2-tramiprosate *in situ* (Patent Owner's assumptions (e) and (g)) to be similarly speculative, and unsupported by evidence. Patent Owner proposes no mechanism by which this might occur, and we note that the site of deuteration of D_2-Val-APS (the third carbon) is not adjacent to, or would be chemically implicated in, the Val-APS to tramiprosate conversion reaction.[26] We find Patent Owner's arguments with respect to proposed substitute claims 21–23 no more persuasive upon repetition, and we incorporate here our findings and conclusions in that respect.

With respect to the unexpected results, as an initial matter, we agree with Patent Owner that Czarnik's compound 6 is not the closest prior art to the claimed compositions. We have found that the closest prior art to

[26] A comparison of compound A2 of Kong (i.e., Val-APS) and tramiprosate indicates that the likely conversion reaction from prodrug to drug is a hydrolysis reaction at the amino group, resulting in carboxylation of the valine residue and creation of an NH_2 terminus on the tramiprosate molecule. Regardless, the deuterated third carbon of D_2-tramiprosate is spatially removed from any such conversion reaction and would be unlikely to play a role in inhibiting the transformation from prodrug to drug.

D_2-Val-APS is Val-APS, as taught by Kong. Similarly, the closest prior art to D_2-tramiprosate is tramiprosate, as also taught by Kong. *See* Section III.J.1, *supra*. As explained, the sum difference between the pairs of compositions is two subatomic particles (neutrons). And as we have also explained, the only correct comparison to demonstrate the increased effectiveness of the prodrug would be a comparison between Val-APS and D_2-Val-APS. Only by such a comparison can the effect of *deuteration* of the prodrug be established. This is science of the most elementary sort, as well as the law of demonstrating unexpected results. *See Baxter Travenol*, 952 F.2d at 392. But Patent Owner offers no evidence of any such comparison.

We have acknowledged that the ’323 patent demonstrates that D_2-tramiprosate demonstrates a 14% increased AUC compared to tramiprosate, but, as we have explained, we find that that difference represents a difference in degree of effect, rather than a difference in kind, and one that a person of skill in the art might reasonable expect, based upon the teachings of the prior art, particularly those of Schofield. Furthermore, even were we to consider the 14% increase in AUC provided by deuteration to be a somewhat unexpected difference in kind, rather than of degree (which we do not), Patent Owner has not persuaded us that that evidence alone would be sufficient to overcome Petitioner’s strong showing of obviousness. *See Newell Cos. v. Kenney Mfg. Co.*, 864 F.2d 757, 769 (Fed. Cir. 1988), *cert. denied*, 493 U.S. 814 (1989) (holding that although the record may establish evidence of secondary considerations which are indicia of nonobviousness, the record may also establish such a strong case of obviousness that the objective evidence of nonobviousness is not sufficient to outweigh the evidence of obviousness); *see also Pfizer Inc. v. Apotex Inc.*, 480 F.3d 1348, 1372 (Fed. Cir. 2007) (unexpected results not sufficient to outweigh a strong showing of obviousness).

E. Conclusion

Having reviewed the parties' arguments, and analyzed the evidence of record, we conclude that the preponderance of the evidence demonstrates that a person of ordinary skill in the art would have found it obvious to combine the cited prior art references cited in Grounds 1–3 to arrive at the proposed substitute claims 21–23 and that Patent Owner's arguments are not sufficient to overcome the preponderance of the evidence of unpatentability.[27]

V. CONCLUSION

For the foregoing reasons, and after having analyzed the entirety of the record and assigning appropriate weight to the cited supporting evidence,

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[27] Should Patent Owner wish to pursue amendment of the challenged claims in a reissue or reexamination proceeding subsequent to the issuance of this decision, we draw Patent Owner's attention to the April 2019 *Notice Regarding Options for Amendments by Patent Owner through Reissue or Reexamination during a Pending AIA Trial Proceeding*. *See* 84 Fed. Reg. 16654 (Apr. 22, 2019). If Patent Owner chooses to file a reissue application or a request for reexamination of the challenged patent, we remind Patent Owner of its continuing obligation to notify the Board of any such related matters through updated mandatory notices. *See* 37 C.F.R. § 42.8(a)(3), (b)(2) (2019).

we determine that Petitioner has established, by a preponderance of the evidence, that claims 1–20 of the ’323 patent are unpatentable.

Additionally, for the foregoing reasons, and after having analyzed the entirety of the record and assigning appropriate weight to the cited supporting evidence, we determine that Petitioner has established, by a preponderance of the evidence, that proposed substitute claims 21–23 are unpatentable, and we consequently deny Patent Owner’s Contingent Motion to Amend.

V. ORDER

Accordingly, it is

ORDERED that Petitioner has shown by a preponderance of the evidence that claims 1–20 of the ’323 Patent are unpatentable;

ORDERED that Petitioner has shown by a preponderance of the evidence that substitute claims 21–23 are unpatentable;

ORDERED that Patent Owner’s Contingent Motion to Amend is denied; and

FURTHER ORDERED that, because this is a Final Written Decision, parties to the proceeding seeking judicial review of the decision must comply with the notice and service requirements of 37 C.F.R. § 90.2.

In summary:

Claims	35 U.S.C. §	Reference(s)/Basis	Claims Shown Unpatentable	Claims Not Shown Unpatentable
1–17, 19, 20	103	Kong, Czarnik	1–17, 19, 20	
1–20	103	Kong, Czarnik, Tolar	1–20	
3, 19	103	Czarnik, Kong	3, 19	
Overall Outcome			1–20	

Motion to Amend Outcome	Claims
Substitute Claims: Proposed in the Amendment	21–23
Substitute Claims: Revised Motion to Amend Granted	
Substitute Claims: Revised Motion to Amend Denied	21–23

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